

Comprehensive Literature Review

Machine learning for alpha signal combination and portfolio construction: what is known, what is solved, and what remains open

Machine Learning for Alpha Signal Combination and Portfolio Construction

This report is part of a pre-empirical research proposal. It distinguishes verified evidence, research inference, proposed design choices and hypotheses. No backtest results are reported; expected outcomes are hypotheses.

Contents

Comprehensive Literature Review: Machine Learning for Alpha Signal Combination and Portfolio Construction	3
Executive Summary	3
1. Scope, Search Protocol and Evidence Standards	4
2. Evolution of the Literature	4
3. Key Papers at a Glance	7
4. Machine-Learning Methods: Where They Add Economic Value	8
5. Portfolio Construction: What the Evidence Says	8
6. Prediction Is Not Investment Value	9
7. What Is Solved and What Remains Open	10
8. The BlackRock "Augmented Investment Management" Framework	10
9. Research Opportunities (summary)	11
Appendix A. Core Literature Matrix (condensed)	12
Appendix B. Literature Universe (200 entries)	18
References	29

Comprehensive Literature Review: Machine Learning for Alpha Signal Combination and Portfolio Construction

Executive Summary

This review covers the published and working-paper literature on using machine learning (ML) to combine many return-predictive signals into portfolios. The literature universe assembled for this project contains more than 180 entries. About 35 are core papers analysed in detail in the literature matrix, and the rest are foundational, methodological, data and inference references. Every entry carries an evidence label; DOIs were checked against Crossref where available (see o8_References).

Five conclusions organise the rest of this document.

1. Prediction is largely solved as an engineering problem, but its economic value is not. Flexible models (trees, neural networks, autoencoders, adversarial SDF estimators) reliably beat linear models on statistical criteria and gross long-short spreads (Gu, Kelly & Xiu, 2020; Gu, Kelly & Xiu, 2021; Chen, Pelger & Zhu, 2024; Freyberger, Neuhierl & Weber, 2020). The most-cited evidence, however, evaluates gross, sort-based portfolios.

2. The implementability reckoning has arrived. ML alpha is concentrated in microcaps, distressed and hard-to-arbitrage stocks and in high limits-to-arbitrage states. It deteriorates under realistic trading costs (Avramov, Cheng & Metzker, 2023). Net performance of common one-month-horizon ML models is near zero after 2004 (Blitz et al., 2023). The average published anomaly earns only a few basis points per month net and post-publication (Chen & Velikov, 2023; Chen & Welch, 2026). Researcher design choices produce non-standard errors larger than standard errors (Chen, Hanauer & Kalsbach, 2024; Lalwani, Meshram & Jindal, 2025; Menkveld et al., 2024). Some sophisticated ML combinations remain profitable after costs (Azevedo, Hoegner & Velikov, 2023).

3. The frontier has moved from "predict better" to "learn under the economic objective". Parametric portfolio policies (Brandt, Santa-Clara & Valkanov, 2009) were extended to transaction costs (DeMiguel et al., 2020), to neural networks (Simon, Weibels & Zimmermann, 2026), and to ML under an implementable-efficient-frontier criterion (Jensen et al., 2026). Learning weights directly under a cost-aware objective dominates predict-then-optimize after costs. Any new paper claiming "cost-aware ML signal combination" must be positioned against these papers.

4. Signal efficacy is time-varying, but evidence of implementable state-dependent combination at the stock level is thin. Factor returns are timeable (Haddad, Kozak & Santosh, 2020; Neuhierl et al., 2024; Kagkadis et al., 2024; Ehsani & Linnainmaa, 2022). Anomalies depend on sentiment and market states (Stambaugh, Yu & Yuan, 2012; Muller & Preissler, 2026), and momentum crashes are predictable (Daniel & Moskowitz, 2016). A conditional multifactor portfolio survives costs at the factor level (DeMiguel, Martin-Utrera & Uppal, 2024). Theory says trading speed and allocation should depend on liquidity regimes (Collin-Dufresne, Daniel & Saglam, 2020). We found no paper that cleanly measures whether state-dependent signal weights add net-of-cost value at the stock level, separately from nonlinearity and from the cost-aware objective (research inference; see the closest-prior-art tracker).

5. Uncertainty and non-stationarity are emerging, not settled. Formal inference for neural-network expected returns and its portfolio use is new (Liao et al., 2025). So is joint selection of training windows and complexity (Capponi et al., 2025), and meta-learning under regime shifts (Wang & Lera, 2025). Learning from forecast errors improves optimiser robustness in low dimensions (Barroso & Saxena, 2022).

The most defensible research opportunity is therefore not another model. It is a rigorous, multiple-testing-robust **attribution of where the net-of-cost value of ML signal combination comes from:** nonlinearity, state dependence, a cost-aware objective and forecast uncertainty, including their interactions. The attribution should come with economic mechanisms. This is developed in the Research Gap report and the Selected Research Design.

1. Scope, Search Protocol and Evidence Standards

Scope. Cross-sectional equity return prediction with many signals; signal/forecast combination; portfolio construction (mean-variance, risk-based, robust, end-to-end, parametric policies); transaction costs and capacity; factor timing and regimes; backtest integrity and inference; interpretability.

Sources. Publisher pages (Oxford Academic, Wiley, Elsevier, INFORMS, Cambridge, Taylor & Francis, Springer), NBER, SSRN, arXiv (only where relevant), authors' repositories, association policy pages, and the Crossref and OpenAlex APIs for bibliographic verification and journal metrics. Blogs and content farms were not used as evidence.

Evidence labels used throughout the package.

Label	Meaning
Verified	Stated in the paper abstract, publisher page, authors' repository, or confirmed via Crossref/OpenAlex
[inference]	Our reading or synthesis; not a claim made by the cited authors
[verify]	Detail we could not confirm from an accessible primary source; check full text before citing
Proposed	Our own research idea or design choice
Hypothesis	A pre-registered expectation to be tested, not a result

Limits of this review. Many full texts sit behind paywalls, so some design details (exact sample splits, cost calibrations) are marked [verify]. Journal Impact Factors from Clarivate could not be verified and are not reported as facts. OpenAlex citation metrics are reported instead, with their date.

2. Evolution of the Literature

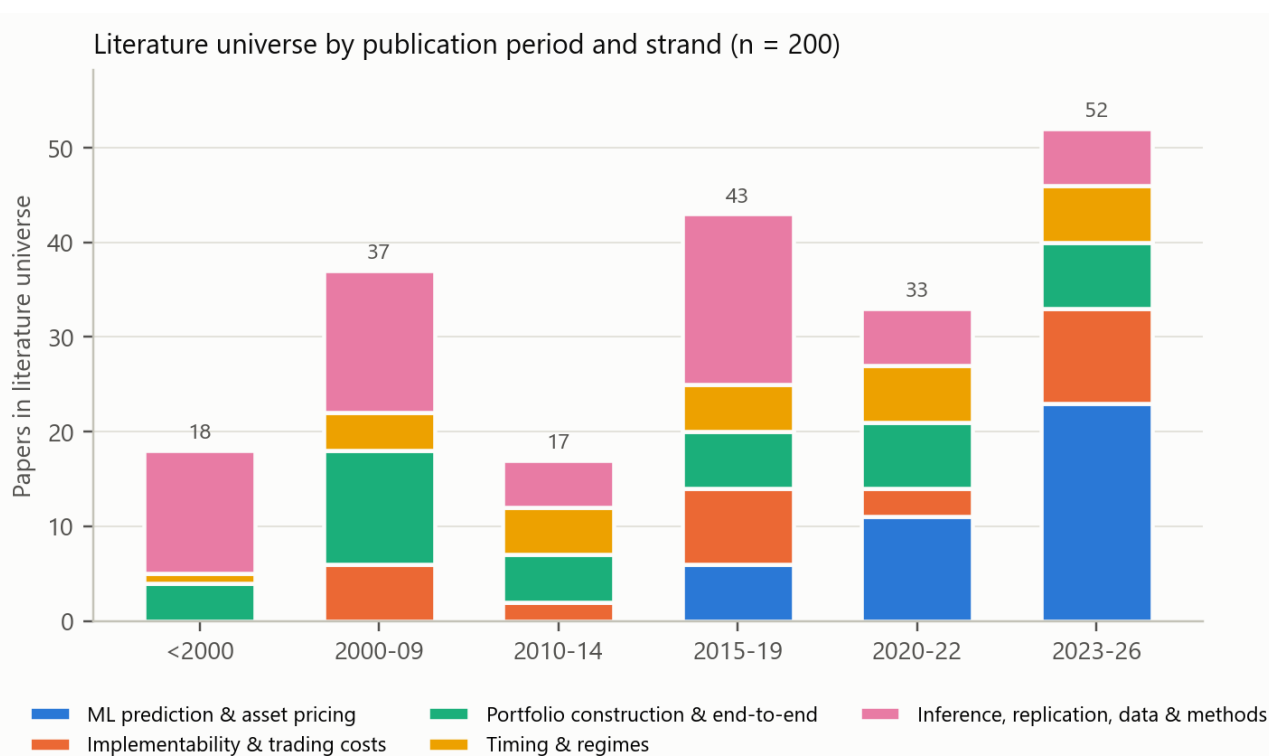


Figure 1. Composition of the assembled literature universe by period and strand (counts from Workbook 1).

2.1 Wave 1: The factor zoo and the multiplicity problem (1990s to 2016)

The characteristic-based literature produced hundreds of predictors. Cochrane's "zoo" framing (Cochrane, 2011) set the question of which characteristics provide independent information. Multiple-testing hurdles rose (Harvey, Liu & Zhu, 2016). Many anomalies failed stricter replication with NYSE breakpoints and value weighting (Hou, Xue & Zhang, 2020), while open replication showed most clearly significant predictors do reproduce (Chen & Zimmermann, 2022; Jensen, Kelly & Pedersen, 2023). Returns decline out of sample and after publication (McLean & Pontiff, 2016), and fewer characteristics provide independent information after 2003 (Green, Hand & Zhang, 2017). This wave supplies both the **signal library** and the **integrity constraints** a modern paper must respect.

2.2 Wave 2: Linear aggregation and shrinkage

Before ML, the canonical combination was the Fama-MacBeth regression with historically averaged slopes (Fama & MacBeth, 1973; Lewellen, 2015). Alternatives included latent-variable/PLS aggregation (Light, Maslov & Rytchkov, 2017) and SDF estimation with economically motivated shrinkage over many anomaly portfolios (Kozak, Nagel & Santosh, 2020). Nonparametric selection showed nonlinearity matters and that many predictors are redundant (Freyberger, Neuhierl & Weber, 2020). In time series, simple forecast combination beats individual models under instability (Rapach, Strauss & Zhou, 2010), a result that makes naive combinations dangerous benchmarks for any "dynamic weighting" claim.

2.3 Wave 3: The ML prediction wave (2018 to present)

Gu, Kelly & Xiu (2020) set the standard design: a pooled panel of stock characteristics, macro interactions and industry dummies; a battery of methods from penalised regressions to neural networks; out-of-sample R^2 and decile spreads. Trees and neural networks won because they capture nonlinear interactions. Extensions added conditional latent factors (Gu, Kelly & Xiu, 2021; Kelly, Pruitt & Su, 2019), adversarial no-arbitrage SDFs with macro states (Chen, Pelger & Zhu, 2024), trees that build test assets (Bryzgalova, Pelger & Zhu, 2025; Cong et al., 2025), principal portfolios using cross-asset signals (Kelly, Malamud & Pedersen, 2023), image-based signals (Jiang, Kelly & Xiu, 2023), and transformers inside the SDF (Kelly et al., 2025). A theoretical "virtue of complexity" argues that heavily parameterised models improve out-of-sample performance (Kelly, Malamud & Zhou, 2024; Didisheim et al., 2023). International evidence confirms broad predictability and the usefulness of combining models (Cakici et al., 2023; Tobek & Hronec, 2021), with market structure mattering (Leippold, Wang & Zhou, 2022).

2.4 Wave 4: The implementability reckoning (2016 to present)

Trading-cost research showed that turnover determines which anomalies survive, and that buy/hold spreads mitigate costs (Novy-Marx & Velikov, 2016; Novy-Marx & Velikov, 2019). Ignoring costs biases model comparisons (Detzel, Novy-Marx & Velikov, 2023). Applied to ML, Avramov, Cheng & Metzker (2023) showed profitability concentrates where arbitrage is hardest, and Blitz et al. (2023) showed the horizon of the training target is a first-order lever for net returns. Chen & Velikov (2023) and Chen & Welch (2026) set a sobering prior for single-signal alpha. Meta-scientific work on non-standard errors (Menkveld et al., 2024) was extended to ML design choices (Chen, Hanauer & Kalsbach, 2024; Lalwani, Meshram & Jindal, 2025). Cost calibration is itself uncertain: live institutional execution costs are "an order of magnitude smaller" than earlier estimates (Frazzini, Israel & Moskowitz, 2018), while fund-based implementation costs appear substantial (Patton & Weller, 2020). A credible paper must therefore treat the cost model as a sensitivity dimension, not a fixed assumption.

2.5 Wave 5: Learning under the economic objective

Modelling weights directly as functions of characteristics (Brandt, Santa-Clara & Valkanov, 2009) was extended to transaction costs, where trade netting makes more characteristics jointly useful (DeMiguel et al., 2020). It was generalised to neural networks (Simon, Weibels & Zimmermann, 2026) and to ML designed for the implementable efficient frontier (Jensen et al., 2026), where learning portfolio weights under a cost-aware objective beats Markowitz-with-ML-forecasts after costs. In OR and ML, decision-focused losses (Elmachtoub & Grigas, 2022), differentiable optimisation layers (Amos & Kolter, 2017; Agrawal et al., 2019) and end-to-end portfolio models (Butler & Kwon, 2023; Costa & Iyengar, 2023; Uysal, Li & Mulvey, 2024; Zhang, Zohren & Roberts, 2020) follow the same logic, but mostly on small asset universes. Reinforcement-learning direct construction reports very high Sharpe ratios in US equities (Cong et al., 2021); independent replication under strict cost assumptions is advisable [inference].

2.6 Parallel strands: timing, regimes, uncertainty, non-stationarity

- **Timing and regimes.** Factor timing (Haddad, Kozak & Santosh, 2020; Neuhierl et al., 2024; Kagkadis et al., 2024), factor momentum (Ehsani & Linnainmaa, 2022), volatility management (Moreira & Muir, 2017; Cederburg et al., 2020; DeMiguel, Martin-Utrera & Uppal, 2024), sentiment-conditional anomalies (Stambaugh, Yu & Yuan, 2012), momentum crashes (Daniel & Moskowitz, 2016; Barroso & Santa-Clara, 2015), state-dependent anomaly performance (Muller & Preissler, 2026), regime-switching allocation (Ang & Bekaert, 2002; Guidolin & Timmermann, 2007; Nystrup, Madsen & Lindstrom, 2018; Shu, Yu & Mulvey, 2025), and liquidity regimes with costs (Collin-Dufresne, Daniel & Saglam, 2020).
- **Uncertainty.** Learning from out-of-sample forecast errors (Barroso & Saxena, 2022); forecast confidence intervals for neural-network expected returns (Liao et al., 2025); robust optimisation (Goldfarb & Iyengar, 2003; Blanchet, Chen & Zhou, 2022); estimation-risk-aware rules (Kan & Zhou, 2007).
- **Non-stationarity.** Joint window/complexity selection (Capponi et al., 2025); meta-learning (Wang & Lera, 2025; Zhao, Kong & Shen, 2023).

3. Key Papers at a Glance

Paper	Core question	Key verified finding	What it leaves open
Gu, Kelly & Xiu (2020)	Which ML measures risk premia best?	Trees and NNs win via nonlinear interactions	Net-of-cost value; implementability
Avramov, Cheng & Metzker (2023)	Does ML survive economic restrictions?	Profits concentrate in hard-to-arbitrage stocks/states; costs erode them	Implementable-by-design combinations
Jensen et al. (2026)	How to use ML for the net frontier?	Learning weights under cost-aware objective dominates	Attribution across nonlinearity, states, uncertainty [inference]
DeMiguel et al. (2020)	How do costs change useful characteristics?	6 to 15 significant characteristics due to netting	Nonlinear/conditional versions
Simon, Weibels & Zimmermann (2026)	Do NN policies beat linear PPP?	75-276 bps/month higher CE, robust to costs	State dependence; decomposition [inference]
Blitz et al. (2023)	Does ML alpha survive costs?	Near-zero net after 2004 for 1-month models; longer horizons help	Signal/state-specific horizons
Azevedo, Hoegner & Velikov (2023)	Expected returns of ML strategies?	Up to 1.42%/month net; LSTM net alpha 1.20% (t=3.46)	Cost-aware learning
Chen, Hanauer & Kalsbach (2024)	Do design choices matter?	1,056 models; returns 0.13%-1.98%/month; NSE 59% above SE	Which choices create net value
Neuhierl et al. (2024)	Can the factor zoo be timed?	Timing improves nearly all factor groups; +20% return for multifactor	Stock-level implementation with costs
DeMiguel, Martin-Utrera & Uppal (2024)	Do conditional multifactor weights survive costs?	Yes, out of sample and net of costs	High-dimensional stock-level conditioning
Collin-Dufresne, Daniel & Saglam (2020)	Optimal allocation under liquidity regimes?	Trading speed higher in persistent, risky, liquid states	Large-scale empirical test
Liao et al. (2025)	How uncertain are NN return forecasts?	Closed-form SEs; uncertainty-averse portfolios improve OOS	Interaction with costs
Chen & Velikov (2023)	Expected anomaly returns net?	About 4 bps/month average net and post-publication	Whether combination rescues value
BlackRock Systematic (2025)	Industrial ML alpha pipeline	Predict-then-optimize with regime resampling and SHAP attribution; no statistical evidence	Everything requiring inference

The full matrix (all fields requested: data, universe, frequency, features, ML, combination, portfolio, benchmark, metrics, period, costs, turnover, risk controls, result, contribution, limitation, what was not solved, extension) is in Workbook 1 (Literature Database).

4. Machine-Learning Methods: Where They Add Economic Value

Method family	Evidence of predictive value	Evidence of net economic value	Assessment for our problem
Penalised linear (ridge, lasso, elastic net, PLS)	Solid, modest (Gu, Kelly & Xiu, 2020; Light, Maslov & Rytchkov, 2017)	Low turnover; strong under costs when used in PPP (DeMiguel et al., 2020)	Mandatory baseline; often hard to beat net
Random forests / GBDT (XGBoost, LightGBM)	Strong, capture interactions (Gu, Kelly & Xiu, 2020; Leung et al., 2021)	Mixed; gains shrink in large caps and after costs (Avramov, Cheng & Metzker, 2023; Blitz et al., 2023)	Main nonlinear workhorse; cheap, interpretable via TreeSHAP
Feed-forward NNs / ensembles	Strong in US panel (Gu, Kelly & Xiu, 2020); large NN portfolios (Simon, Weibels & Zimmermann, 2026)	Positive when trained on economic objective (Simon, Weibels & Zimmermann, 2026; Jensen et al., 2026)	Include as ensemble with seeds; report dispersion
Random features / high complexity	Theory and evidence for complexity (Kelly, Malamud & Zhou, 2024; Didisheim et al., 2023)	JKMP uses random-feature regressions inside a cost-aware design (Jensen et al., 2026)	Useful for economic-loss learner
Autoencoders / latent factors (IPCA, CA)	Good pricing performance (Kelly, Pruitt & Su, 2019; Gu, Kelly & Xiu, 2021)	Weakened under restrictions (Avramov, Cheng & Metzker, 2023)	Risk-model / benchmark role
RNN / LSTM	Useful for sequences and macro states (Chen, Pelger & Zhu, 2024; Azevedo, Hoegner & Velikov, 2023)	LSTM combination profitable net in one study (Azevedo, Hoegner & Velikov, 2023)	Optional; high tuning burden
CNN (images)	Price-pattern signals (Jiang, Kelly & Xiu, 2023)	Signal generation, fast decay likely [inference]	Out of scope (signal generation)
Transformers / attention	Cross-asset information sharing (Kelly et al., 2025; Cong et al., 2021)	Unproven under strict costs [inference]	Robustness extension only
Reinforcement learning	Direct construction (Cong et al., 2021; Moody & Saffell, 2001)	Claims large; replication risk	Not a primary method; high overfitting risk
Mixture-of-experts / gating	Classic architecture (Jacobs et al., 1991)	Little finance evidence net of costs [inference]	Natural way to parameterise state dependence
Meta-/online learning	Regime adaptation (Wang & Lera, 2025; Zhao, Kong & Shen, 2023)	Net value untested [inference]	Robustness dimension
Bayesian / uncertainty-aware	Forecast SEs (Liao et al., 2025); deep ensembles (Lakshminarayanan, Pritzel & Blundell, 2017)	Emerging (Liao et al., 2025; Barroso & Saxena, 2022)	One factor in our design
Graph neural networks	Economic links predict returns (Cohen & Frazzini, 2008)	Data/PIT hard; unproven net [inference]	Out of scope for main paper

Assessment [\[inference\]](#). The binding constraint on economic value is not model expressiveness but the mapping from forecasts to trades: turnover, concentration in illiquid names, and misalignment between the training loss and the investor objective. Additional architectural complexity is justified only if it survives the cost-aware objective and data-snooping corrections.

5. Portfolio Construction: What the Evidence Says

- **Estimation error dominates naive optimisation.** None of 14 sample-based rules consistently beat 1/N out of sample (DeMiguel, Garlappi & Uppal, 2009). Constraints act as shrinkage (Jagannathan & Ma, 2003); shrinkage covariance estimators help in high dimensions (Ledoit & Wolf, 2004; Ledoit & Wolf, 2017). Hierarchical risk parity avoids inversion (Lopez de Prado, 2016).
- **Risk-based portfolios** (minimum variance, risk parity, maximum diversification, HRP) are useful risk baselines (Maillard, Roncalli & Teiletche, 2010; Choueifaty & Coignard, 2008) but ignore alpha. For a signal-combination paper they belong in the risk-model comparison, not as alpha benchmarks [\[inference\]](#).
- **Blending forecasts with confidence.** Black-Litterman (Black & Litterman, 1992) and robust/DRO formulations (Goldfarb & Iyengar, 2003; Delage & Ye, 2010; Blanchet, Chen & Zhou, 2022) offer principled ways to temper ML alphas by their uncertainty.
- **Transaction-cost-aware dynamic trading.** Aim portfolios with partial adjustment (Garleanu & Pedersen, 2013), regime-dependent trading speed (Collin-Dufresne, Daniel & Saglam, 2020), and utility-maximising parametric policies with costs (DeMiguel et al., 2020; Jensen et al., 2026) are the state of the art. Heuristic mitigation (buy/hold bands) is a strong simple benchmark (Novy-Marx & Velikov, 2016).

- **Alignment.** Differences between the alpha model, risk model and constraints create unintended exposures and underestimated risk (Ceria, Saxena & Stubbs, 2012; Lee & Stefek, 2008). This is likely worse for nonlinear ML alphas [hypothesis].
- **End-to-end and decision-focused learning** reduces the gap between loss and objective (Elmachtoub & Grigas, 2022; Butler & Kwon, 2023), but evidence at stock-universe scale with realistic constraints remains limited, except for parametric/economic-loss approaches (Simon, Weibels & Zimmermann, 2026; Jensen et al., 2026).

6. Prediction Is Not Investment Value

Conceptual chain from prediction to investment value (each link is a measured quantity in our design)

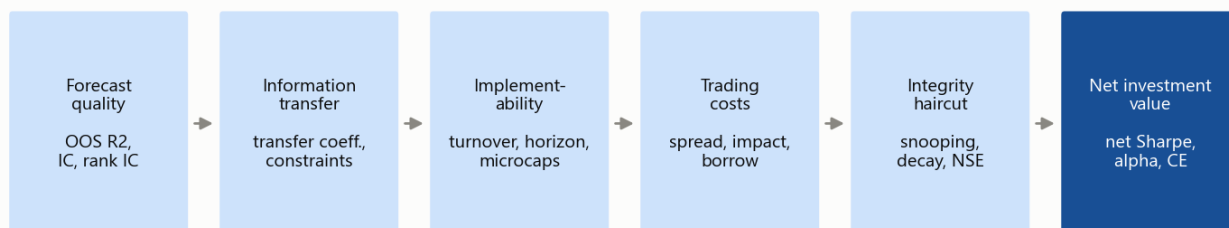


Figure 2. Conceptual chain from forecast quality to net investment value; each link is measured in the proposed design.

Many ML finance papers report strong predictive statistics but weak tradable performance. The mechanisms documented in the literature are:

1. **Tiny R^2 , concentrated where it cannot be traded.** Monthly stock-level R^2 is fractions of a percent, even for the best models (Gu, Kelly & Xiu, 2020). The predictive content concentrates in microcaps, distressed and high-idiosyncratic-volatility stocks (Avramov, Cheng & Metzker, 2023). Equal-weighted sorts amplify it (Hou, Xue & Zhang, 2020).
2. **Loss/objective mismatch.** MSE weights all stocks and errors equally; the investor cares about ranking at the extremes, covariance, and trade costs. Decision-focused and economic-objective learning exist precisely because of this gap (Elmachtoub & Grigas, 2022; Jensen et al., 2026).
3. **Turnover.** One-month-horizon forecasts from fast signals such as reversal, short-term momentum and liquidity shocks create high turnover. Anomalies with more than 50% monthly turnover rarely survive (Novy-Marx & Velikov, 2016), and longer training horizons restore net alpha (Blitz et al., 2023).
4. **The IC-to-IR chain.** The fundamental law links IR to IC and breadth (Grinold, 1989). Constraints reduce the transfer coefficient, so a higher IC can translate into little additional IR (Clarke, de Silva & Thorley, 2002).
5. **Short-leg dependence.** A large share of anomaly profits comes from the short side and bad states (Stambaugh, Yu & Yuan, 2012; Muller & Preissler, 2026), where borrow costs and constraints bind.
6. **Error maximisation.** Plugging noisy ML alphas into mean-variance optimisers amplifies errors (Michaud, 1989; DeMiguel, Garlappi & Uppal, 2009).
7. **Decay and publication.** Returns fall out of sample and after publication (McLean & Pontiff, 2016). Recent net alpha of published anomalies is near zero (Chen & Welch, 2026).
8. **Researcher degrees of freedom.** Non-standard errors from design choices rival or exceed standard errors (Chen, Hanauer & Kalsbach, 2024; Lalwani, Meshram & Jindal, 2025; Menkveld et al., 2024). Multiple testing inflates backtests (Harvey, Liu & Zhu, 2016; Bailey & Lopez de Prado, 2014; Bailey et al., 2017).
9. **Non-stationarity.** Relationships shift; longer windows help complex models but include stale regimes (Capponi et al., 2025).

Implication for our research. Every experiment must report prediction metrics (OOS R^2 , IC, rank IC, ICIR), portfolio metrics (gross and net Sharpe, Sortino, drawdown, turnover, alpha, capacity), and the **conversion rates** between them. These are, for example, net Sharpe per unit of IC and cost drag per unit of turnover. This makes "prediction vs. investment value" a measured result rather than a caveat.

7. What Is Solved and What Remains Open

Largely solved (do not claim novelty here):

- ML beats linear models on statistical accuracy and gross spreads in US and global equities.
- Trees and NNs capture nonlinear interactions; price-trend, liquidity and volatility characteristics dominate predictions.
- Costs matter first-order; turnover and horizon are key levers; simple mitigation helps.
- Learning portfolio weights under a cost-aware economic objective beats predict-then-optimize net of costs (JKMP2026; DMNU2020 linear; SWZ2023 nonlinear).
- Factor returns are timeable at the factor-portfolio level.
- Backtest-integrity tools exist (DSR, PBO, SPA, MCS, FDR).

Partially solved:

- State-conditional combination net of costs: shown for low-dimensional factor sets (DMU2024), not stock-level high-dimensional signals.
- Uncertainty-aware ML portfolios: new working-paper evidence (LMNS2025), interaction with costs untested.
- Design-choice robustness: documented (CHK2024, LAL2025), but the economic sources of dispersion are not identified.

Open (to the best of our search; re-check before submission):

- A unified attribution of net-of-cost ML value to nonlinearity, state dependence, economic objective and uncertainty, including their interactions.
- Whether conditioning helps only when costs are internalised (the interaction), and whether the realised state dependence matches economic theory (DM2016, SY2012, CDS2020).
- Whether conclusions survive formal data-snooping control across all tried configurations.

8. The BlackRock "Augmented Investment Management" Framework

What it proposes (verified from the white paper, BlackRock Systematic, November 2025, professional-investor material). AIM, started in 2014, reframes signal combination as forecasting excess returns ("alpha forecasts") from a large signal library, which BlackRock's Figure 1 shows growing to roughly 1,200 stock-selection signals by 2025. The pipeline has four stages: feature processing (outliers, imputation, time-series transforms, optional risk-factor neutralisation); ML training (regularised linear models, gradient-boosting ensembles, neural networks, proprietary variants) chosen by a large-scale configuration search with cross-validation; prediction post-processing; and a separate optimiser and simulator that accounts for transaction costs, borrow costs and fund constraints. Model parameters are tuned to maximise a metric such as the information ratio. Training data typically span 10-20 years. Interpretability uses Shapley-regression additive attribution. Regime-awareness is achieved by resampling so that negative regimes are over-represented. The illustrative defensive model had better drawdown behaviour but about a 20% overall performance haircut. The authors list non-stationarity (signal decay, cyclical shifts) as ongoing research, pointing to adaptive/online learning, transfer learning and domain adaptation.

Assumptions [inference]. (i) Separation of forecasting from portfolio construction is acceptable, with costs handled only in the optimiser and simulator. (ii) Cross-validation on financial time series controls overfitting, although standard k-fold without purging/embargo can leak. (iii) Regime robustness can be induced by reweighting history rather than by conditioning on observable states. (iv) Tuning on backtest IR across a large configuration search does not create selection bias.

What later academic research improves. Cost-aware learning of weights instead of pure predict-then-optimize (Jensen et al., 2026; DeMiguel et al., 2020; Simon, Weibels & Zimmermann, 2026). Purged/embargoed validation (Lopez de Prado, 2018). Deflated Sharpe and PBO for configuration search (Bailey & Lopez de Prado, 2014; Bailey et al., 2017). State-conditional combination that survives costs (DeMiguel, Martin-Utrera & Uppal, 2024). Forecast uncertainty (Liao et al., 2025). Horizon choice (Blitz et al., 2023).

What remains unanswered. The white paper provides no out-of-sample statistical tests, no net-of-cost performance tables, no turnover or capacity figures, and no attribution of value to its components. Its data are proprietary.

What we do differently (proposed). We treat each AIM design component as a factor in a controlled experiment on public, reproducible data: nonlinearity, conditioning (versus resampling), cost-awareness in training (versus only in the optimiser), and uncertainty. We use one optimiser and one cost model with sensitivity analysis, log every configuration for data-snooping control, and test whether the realised state dependence matches economic theory.

9. Research Opportunities (summary)

1. Attribution of net-of-cost ML value across nonlinearity, conditioning, objective and uncertainty (selected; see 05).
2. Forecasting signal efficacy with signal-level meta-features and implementing it at stock level with netting.
3. Uncertainty-aware alpha combination and robust cost-aware optimisation.
4. Scalable differentiable cost-aware optimisation layers.
5. Regime-gated mixture-of-experts combination with economically interpretable gates.

The full 15-idea matrix and five concept proposals are in 04_Paper_Concepts and Workbook 3.

Appendix A. Core Literature Matrix (condensed)

Full fields for every paper are in Workbook 1. Text is truncated here for layout.

Paper	Venue	Data / universe	Method	Transaction costs	Main result	Not solved
Garleanu & Pedersen (2013)	Journal of Finance 68(6):2309-2340	Commodity futures application [verify] ; Futures	ML: n/a / Portfolio: Closed-form: trade partially toward an aim portfolio that is itself a weighted average of current and expected future Markowitz portfolios	Quadratic costs, proportional to risk	Two principles: aim in front of the target and trade partially toward it.	Learning signal decay and nonlinear signal interactions from data.
Bailey & Lopez de Prado (2014)	Journal of Portfolio Management 40(5):94-107	n/a; n/a	ML: n/a / Portfolio: n/a	n/a	DSR corrects for selection bias under multiple testing and non-normal returns.	n/a
Lewellen (2015)	Critical Finance Review 4(1):1-44	US equities [verify] ; All stocks, excluding tiny stocks, large stocks	ML: Fama-MacBeth cross-sectional regressions with rolling slope averages / Portfolio: Sorts on fitted expected returns	Not central	Fitted expected returns from rolling FM slopes have strong out-of-sample predictive power.	Costs, nonlinearity.
Novy-Marx & Velikov (2016)	Review of Financial Studies 29(1):104-147	US anomalies with effective spread estimates; US stocks	ML: n/a / Portfolio: Long-short with buy/hold spreads	Effective bid-ask spreads	Buy/hold spreads are the most effective mitigation; most anomalies with under 50% monthly turnover earn significant net spreads, few with higher turnover do; size, value and profitability strategies have the most capacity.	Optimised cost-aware combination.
Gu, Kelly & Xiu (2020)	Review of Financial Studies 33(5):2223-2273	CRSP/Compustat US equities, monthly, 1957-2016 (NBER w25398); All US common stocks (roughly 30,000 over the sample) [verify exact filters]	ML: OLS, OLS-3, PLS, PCR, elastic net, generalized linear model with group lasso, random forest, gradient-boosted trees, neural networks with 1-5 hidden layers / Portfolio: Decile sorts...	Not modelled in the headline portfolio results [inference from design; verify appendix]	Trees and neural networks perform best; gains come from nonlinear predictor interactions; price trends, liquidity and volatility are the dominant predictors.	Whether the predictive gains survive trading costs, implementable constraints and realistic capacity.
DeMiguel et al. (2020)	Review of Financial Studies 33(5):2180-2222	US equities with a large set of firm characteristics [verify count and dates] ; US common stocks	ML: Parametric portfolio policy (linear in characteristics) with regularisation / Portfolio: Parametric portfolio policy with transaction costs	Explicit, inside the objective	Transaction costs increase the number of jointly significant characteristics from 6 to 15, because trades implied by different characteristics partly cancel out (netting).	Nonlinear and state-dependent combinations under costs.
Collin-Dufresne, Daniel & Saglam (2020)	Journal of Financial Economics 136(2):379-406	Model with calibrated application [verify] ; Multi-asset	ML: n/a / Portfolio: Aim portfolio = weighted average of conditional MV portfolios across future states	Regime-dependent quadratic costs	Trading speed should vary with the regime; it can be optimal to overweight low-Sharpe assets that stay liquid in crises.	Empirical, high-dimensional, learned version for stock-level signals.

Paper	Venue	Data / universe	Method	Transaction costs	Main result	Not solved
Chen & Zimmermann (2022)	Critical Finance Review 11(2):207-264	319 characteristics in the paper; website provides 209 firm-level signals (212 incl. 3 CRSP-based) through Dec 2024; US stocks	ML: n/a / Portfolio: Original-paper long-short portfolios	Not part of the paper [verify]	98% of clearly significant predictors reproduce with $t > 1.96$; slope of reproduced on original t-stats is 0.88 with R^2 of 82%.	n/a
Avramov, Cheng & Metzker (2023)	Management Science 69(5):2587-2619	US equities; ML models from the recent literature (neural networks, conditional autoencoder, IPCA, adversarial SDF) [verify full list] ; US common...	ML: Deep learning / latent-factor ML models / Portfolio: Long-short and tangency portfolios implied by the models	Reasonable trading costs applied; performance deteriorates because of high turnover and extreme positions	ML profitability comes from difficult t-to-arbitrage stocks and high limits-to-arbitrage states; excluding microcaps, distressed stocks or high-volatility episodes considerably attenuates it; costs erode it further.	How to build ML signal combinations whose value survives restrictions and costs.
Blitz et al. (2023)	Journal of Financial Data Science 5(4):40-65	US equities [verify] ; [verify]	ML: Common ML models (trees, neural networks) [verify] / Portfolio: Long-short portfolios with efficient trading rules	Explicit	Common ML models have high full-sample gross alphas but net performance near zero after 2004; training on longer horizons (3, 6, 12 months) and efficient trading rules yields significantly positive net returns.	Signal-specific and state-specific horizons.
Azevedo, Hoegner & Velikov (2023)	Working paper (SSRN 4702406); AFA programme	US anomaly characteristics; US stocks	ML: Range from simple to sophisticated ML, including LSTM / Portfolio: Long-short, value-weighted variants	Explicit effective-spread costs	Sophisticated ML strategies earn net out-of-sample returns up to 1.42% per month; an LSTM combination has a six-factor generalized net alpha of 1.20% ($t = 3.46$).	Cost-aware learning and state dependence.
Chen & Velikov (2023)	Journal of Financial and Quantitative Analysis...	204 anomalies; effective bid-ask spreads; US stocks	ML: n/a / Portfolio: Long-short with cost mitigation	Effective bid-ask spreads (high- and low-frequency estimates)	Net of these effects, the average anomaly earns about 4 bps per month; cost mitigation works pre-publication but net returns are tiny afterward.	Whether combination/netting rescues the value.
Jensen, Kelly & Pedersen (2023)	Journal of Finance 78(5):2465-2518	153 factors, 93 countries (jpkfactors.com; CC BY-NC 4.0; updated through Dec 2025); Global	ML: Bayesian hierarchical model of factor alphas / Portfolio: Factor portfolios; tangency portfolio	Not central	Most factors replicate, cluster into 13 themes (most in the tangency portfolio), work out of sample across 93 countries, and evidence is strengthened by the number of factors.	Implementable combination.

Paper	Venue	Data / universe	Method	Transaction costs	Main result	Not solved
Neuhierl et al. (2024)	Working paper (SSRN 4376898); AFA/AEA conference programmes	300+ equity risk factors and a high-dimensional predictor set; US factor portfolios	ML: Partial least squares and other dimension-reduction/shrinkage methods / Portfolio: Timed multifactor portfolio	[verify treatment]	Almost all factor groups can be timed, most for profitability and value; past factor returns and volatility are the best single predictors but PLS aggregation dominates; timing raises the multifactor portfolio return by about 20%.	Whether factor-timing information survives when combined at the stock level inside a cost-aware optimizer [inference].
DeMiguel, Martin-Utrera & Uppal (2024)	Journal of Finance 79:3859-3891	US factor portfolios; replication package on Mendeley Data; Factor portfolios	ML: n/a / Portfolio: Conditional multifactor portfolio	Explicit	Single-factor volatility management does not survive costs, but a conditional multifactor portfolio outperforms its unconditional counterpart out of sample and net of costs; factor risk prices generally fall with market volatility.	Stock-level, high-dimensional, nonlinear conditioning.
Chen, Hanauer & Kalsbach (2024)	Working paper (SSRN 5031755)	US equities with anomaly inputs; US stocks, with/without microcaps	ML: Multiple ML models across 1,056 design combinations / Portfolio: Top-minus-bottom portfolios	[verify]	Seven design choices (model, target, target transformation, pre-publication inputs, feature compression, rolling vs expanding window, microcaps) generate monthly top-minus-bottom returns from 0.13% to 1.98%; the non-standard error exceeds the standard...	Which choices create net-of-cost value and why.
Liao et al. (2025)	Working paper (arXiv 2503.00549)	Equity panel [verify]; [verify]	ML: Neural networks; closed-form and bootstrap standard errors / Portfolio: Uncertainty-averse investment framework (shrinkage interpretation)	[verify]	NN return forecasts share the asymptotic distribution of classic nonparametric estimators, giving closed-form standard errors; using the intervals improves out-of-sample performance.	Interaction of uncertainty with trading costs and state dependence.

Paper	Venue	Data / universe	Method	Transaction costs	Main result	Not solved
Jensen et al. (2026)	Review of Financial Studies (published online 2026)	JKP global characteristics dataset (US monthly characteristics, daily and monthly returns); WRDS access required (code README); US stocks	ML: Ridge-type regressions on random Fourier features; separate models for returns at horizons $t+1..t+12$ (code README) / Portfolio: Trading-cost-aware optimisation; methods compared:...	Explicit trading-cost model entering the optimisation (price impact increasing in trade size relative to...	After costs, Markowitz-ML and sorts have downward-bending frontiers; Static-ML earns positive net Sharpe but negative utility unless tuned; Portfolio-ML, which learns weights under an economic objective, works best out of sample.	[inference] How much of the net gain comes from nonlinearity versus conditioning versus the objective; whether conditioning on market states adds value once costs are...
Simon, Weibels & Zimmermann (2026)	Management Science (2026; Crossref record) - earlier CFR...	US equities with firm characteristics [verify]; US stocks	ML: Feed-forward neural networks mapping characteristics to weight tilts / Portfolio: Deep parametric portfolio policy with weight/leverage constraints	Incorporated; results robust after costs [verify whether costs enter training]	Network-based policies achieve 75-276 bps higher monthly certainty-equivalent returns than a comparable linear policy, robust to costs and short-sale restrictions.	Decomposition of gains; state conditioning.
Brandt, Santa-Clara & Valkanov (2009)	Review of Financial Studies 22(9):3411-3447	US equities [verify dates]; US common stocks	ML: Parametric (linear) weight function estimated by maximising average utility / Portfolio: Benchmark weights plus characteristic tilts	Extension discussed [verify]	Direct weight modelling scales to large cross-sections and delivers economically large gains.	Nonlinearity, many signals, costs, state dependence.
Rapach, Strauss & Zhou (2010)	Review of Financial Studies 23(2):821-862	US equity premium with Welch-Goyal predictors; Market index	ML: Forecast combination (mean, median, trimmed, discount-MSFE) / Portfolio: Asset allocation utility gains	n/a	Combination forecasts deliver consistent out-of-sample gains because model uncertainty and instability impair individual models.	Cross-sectional signal combination.
McLean & Pontiff (2016)	Journal of Finance 71(1):5-32	97 predictors; US stocks	ML: n/a / Portfolio: Long-short	n/a	Portfolio returns are 26% lower out of sample and 58% lower post-publication.	Decay-aware combination.
Frazzini, Israel & Moskowitz (2018)	Working paper (SSRN 3229719)	\$1.7 trillion of live trade execution data from a large institutional manager, 21 developed markets, 19 years; Developed equity markets	ML: n/a / Portfolio: n/a	Measured directly	Actual trading costs are an order of magnitude smaller than previous studies suggest.	n/a
Haddad, Kozak & Santosh (2020)	Review of Financial Studies 33(5):1980-2018	US anomaly portfolios; Anomaly long-short portfolios	ML: PCA + predictive regressions with shrinkage / Portfolio: Optimal factor-timing portfolio	[verify]	Market-neutral factors are strongly and robustly predictable; timing substantially improves performance relative to static factor investing.	Stock-level implementation of timed signals.

Paper	Venue	Data / universe	Method	Transaction costs	Main result	Not solved
Tobek & Hronec (2021)	Journal of Financial Markets 56:100588	153 anomalies, global equities; US and international; liquid universe	ML: ML aggregation into a single mispricing signal / Portfolio: Long-short	Liquid universe	ML aggregation of all anomalies is profitable around the globe and survives in a liquid universe; non-US past performance does not help pick US winners.	Cost-aware and state-aware combination.
Barroso & Saxena (2022)	Review of Financial Studies 35(3):1 222-1278	Portfolio datasets; replication data on Harvard Dataverse; [verify]	ML: Empirical-Bayes-style learning from forecast errors / Portfolio: Mean-variance with calibrated inputs	[verify]	Forecast errors have predictable patterns; learning from them yields portfolios whose performance meets expectations.	Stock-level ML alpha uncertainty.
Chen, Pelger & Zhu (2024)	Management Science 70(2):714-750	US stocks with firm characteristics and many macroeconomic time series; US stocks	ML: Adversarial (GAN-style) networks; recurrent networks for macro states / Portfolio: SDF portfolio	Not central	No-arbitrage criterion, adversarial test assets and macro states improve SDF estimation.	Net-of-cost value of state conditioning.
Lalwani, Meshram & Jindal (2025)	European Financial Management 32(3):816-833	Equity panel [verify] [market and dates] ; [verify]	ML: Seven ML models / Portfolio: 5,376 portfolios across design choices	Considered	Non-standard errors are up to five times the standard errors; about one-third of portfolios earn significant returns after costs.	Economic source of the dispersion.
Capponi et al. (2025)	Working paper (arXiv 2512.23596, revised 2026)	US equity industry portfolios, three decades; Industry portfolios	ML: Adaptive joint selection of complexity and window with guarantees / Portfolio: [verify]	n/a	Adaptive selection improves out-of-sample R2 on industry portfolios by 14% relative to fixed-window and regime-switching benchmarks, with large gains in recessions.	Economic value net of costs.
Wang & Lera (2025)	Journal of Financial Markets 79:101042	Two major equity markets [verify] ; [verify]	ML: FinPFN: finance-specific prior-data fitted network (meta-learning) / Portfolio: [verify]	[verify]	Superior performance relative to ML benchmarks, especially in turbulent periods.	Net-of-cost value of adaptation.
Muller & Preissler (2026)	Journal of Banking & Finance 190:107746	57 countries, 1980-2019 (SSRN version); International stocks	ML: n/a / Portfolio: Anomaly long-short	[verify]	Anomalies perform better in unfavourable market states, driven mostly by the short side (reported magnitudes differ between SSRN and journal versions; verify).	Exploiting state dependence implementably.

Paper	Venue	Data / universe	Method	Transaction costs	Main result	Not solved
Cakici et al. (2023)	Journal of Economic Dynamics and Control 155:104725	46 countries, 148 firm characteristics; International stocks	ML: Repertoire of ML models and their combination / Portfolio: Long-short	[verify]	Predictability comes mainly from momentum, reversal, value and size; combining models is particularly effective; performance depends on firm size, recency of information, market breadth and idiosyncratic risk.	Implementable combination.
Kelly, Malamud & Zhou (2024)	Journal of Finance 79(1):459-503	US market return with standard predictors; Market index (timing)	ML: Random Fourier features ridge regression / Portfolio: Market-timing strategy	[verify]	Complex models dominate simple ones in theory and US market timing; strategies divest before recessions (14 of 15 out of sample).	Complexity vs costs in cross-sectional combination.
BlackRock Systematic (2025)	Industry white paper (BlackRock, Nov 2025; for...	Proprietary signal library (~1,200 stock-selection signals by 2025 per Figure 1); 10-20 years of training data; Illustrations on developed-market...	ML: Regularised linear models, gradient boosting ensembles, neural networks, proprietary variants; large configuration search; cross-validation / Portfolio: Separate high-fidelity...	Inside the performance simulator and optimizer; details not disclosed	Qualitative framework; illustrative defensive model trained on resampled negative regimes had lower drawdowns but about a 20% performance haircut; Shapley-based attribution reveals a nonlinear volatility effect.	Whether ML combination adds value net of costs, which component creates it, and how to adapt to signal decay (listed by the authors as ongoing research:...

Appendix B. Literature Universe (200 entries)

Sorted by strand and year. Relevance: 1-5 (team judgement).

ID	Citation	Title	Venue	Strand	Rel.
BLLP2025	Bryzgalo va et al. (2025)	Missing Financial Data	Review of Financial Studies 38(3):803-882	Data & measurement	4
FHNW2025	Freyberg er et al. (2025)	Missing Data in Asset Pricing Panels	Review of Financial Studies 38(3):760-802	Data & measurement	4
CZ2022	Chen & Zimmer mann (2022)	Open Source Cross-Sectional Asset Pricing	Critical Finance Review 11(2):207-264	Data & measurement	5
SY2017	Stambau gh & Yuan (2017)	Mispricing Factors	Review of Financial Studies 30(4):1270-1315	Data & measurement	2
FF2015	Fama & French (2015)	A Five-Factor Asset Pricing Model	Journal of Financial Economics 116(1):1-22	Data & measurement	3
HXZ2015	Hou, Xue & Zhang (2015)	Digesting Anomalies: An Investment Approach	Review of Financial Studies 28(3):650-705	Data & measurement	3
FP2014	Frazzini & Pedersen (2014)	Betting Against Beta	Journal of Financial Economics 111(1):1-25	Data & measurement	2
AMP2013	Asness, Moskowi tz & Pedersen (2013)	Value and Momentum Everywhere	Journal of Finance 68(3):929-985	Data & measurement	2
CF2008	Cohen & Frazzini (2008)	Economic Links and Predictable Returns	Journal of Finance 63(4):1977-2011	Data & measurement	2
AHXZ2006	Ang et al. (2006)	The Cross-Section of Volatility and Expected Returns	Journal of Finance 61(1):259-299	Data & measurement	2
BW2006	Baker & Wurgler (2006)	Investor Sentiment and the Cross-Section of Stock Returns	Journal of Finance 61(4):1645-1680	Data & measurement	3
PS2003	Pastor & Stambau gh (2003)	Liquidity Risk and Expected Stock Returns	Journal of Political Economy 111(3):642-685	Data & measurement	3
CAR1997	Carhart (1997)	On Persistence in Mutual Fund Performance	Journal of Finance 52(1):57-82	Data & measurement	2
SHU1997	Shumwa y (1997)	The Delisting Bias in CRSP Data	Journal of Finance 52(1):327-340	Data & measurement	4
FF1993	Fama & French (1993)	Common Risk Factors in the Returns on Stocks and Bonds	Journal of Financial Economics 33(1):3-56	Data & measurement	3
SPO2026	Wang & Hasuike (2026)	Decision-Induced Ranking Explains Prediction Inflation and Excessive Turnover in...	Working paper (arXiv 2605.01176)	End-to-end / decision-focused	4
SWZ2023	Simon, Weibels & Zimme rmann (2026)	Deep Parametric Portfolio Policies	Management Science (2026; Crossref record) -...	End-to-end / decision-focused	5

ID	Citation	Title	Venue	Strand	Rel.
ULM2024	Uysal, Li & Mulvey (2024)	End-to-End Risk Budgeting Portfolio Optimization with Neural Networks	Annals of Operations Research 339(1-2):397-426	End-to-end / decision-focused	3
BK2023	Butler & Kwon (2023)	Integrating Prediction in Mean-Variance Portfolio Optimization	Quantitative Finance 23(3):429-452	End-to-end / decision-focused	4
CI2023	Costa & Iyengar (2023)	Distributionally Robust End-to-End Portfolio Construction	Quantitative Finance 23(10):1465-1482	End-to-end / decision-focused	3
EG2022	Elmachtoub & Grigas (2022)	Smart 'Predict, then Optimize'	Management Science 68(1):9-26	End-to-end / decision-focused	4
WGRZ2022	Wood et al. (2022)	Trading with the Momentum Transformer: An Intelligent and Interpretable Architecture	arXiv 2112.08534 [verify journal version]	End-to-end / decision-focused	2
CTWZ2021	Cong et al. (2021)	AlphaPortfolio: Direct Construction through Deep Reinforcement Learning and...	Working paper (SSRN 3554486; NBER w35195)	End-to-end / decision-focused	4
ZZR2020	Zhang, Zohren & Roberts (2020)	Deep Learning for Portfolio Optimization	Journal of Financial Data Science 2(4):8-20	End-to-end / decision-focused	2
AAB2019	Agrawal et al. (2019)	Differentiable Convex Optimization Layers	Advances in Neural Information Processing Systems 32	End-to-end / decision-focused	3
LZR2019	Lim, Zohren & Roberts (2019)	Enhancing Time-Series Momentum Strategies Using Deep Neural Networks	Journal of Financial Data Science 1(4):19-38	End-to-end / decision-focused	2
AK2017	Amos & Kolter (2017)	OptNet: Differentiable Optimization as a Layer in Neural Networks	Proceedings of ICML 2017 (PMLR 70)	End-to-end / decision-focused	3
DAK2017	Donti, Amos & Kolter (2017)	Task-based End-to-end Model Learning in Stochastic Optimization	Advances in Neural Information Processing Systems 30	End-to-end / decision-focused	3
MS2001	Moody & Saffell (2001)	Learning to Trade via Direct Reinforcement	IEEE Transactions on Neural Networks 12(4):875-889	End-to-end / decision-focused	2
MPR2026	Muller & Preissler (2026)	In Good and in Bad Times? The Relation between Anomaly Returns and Market States	Journal of Banking & Finance 190:107746	Factor timing & regimes	4
FRLUX2025	Zhang (2025)	FR-LUX: Friction-Aware, Regime-Conditioned Policy Optimization for Implementable...	Working paper (arXiv 2510.02986)	Factor timing & regimes	5
SYM2025	Shu, Yu & Mulvey (2025)	Dynamic Asset Allocation with Asset-Specific Regime Forecasts	Annals of Operations Research 346(1):285-318	Factor timing & regimes	3
DMU2024	DeMiguel, Martin-Utrera & Uppal (2024)	A Multifactor Perspective on Volatility-Managed Portfolios	Journal of Finance 79:3859-3891	Factor timing & regimes	5
KNNV2024	Kagkadis et al. (2024)	Factor Timing with Portfolio Characteristics	Review of Asset Pricing Studies 14(1):84-118	Factor timing & regimes	3
NRRZ2024	Neuhierl et al. (2024)	Timing the Factor Zoo	Working paper (SSRN 4376898); AFA/AEA conference...	Factor timing & regimes	5

ID	Citation	Title	Venue	Strand	Rel.
DLRZ2022	Dong et al. (2022)	Anomalies and the Expected Market Return	Journal of Finance 77(1):639-681	Factor timing & regimes	2
EL2022	Ehsani & Linnainmaa (2022)	Factor Momentum and the Momentum Factor	Journal of Finance 77(3):1877-1919	Factor timing & regimes	3
CDS2020	Collin-Dufresne, Daniel & Saglam (2020)	Liquidity Regimes and Optimal Dynamic Asset Allocation	Journal of Financial Economics 136(2):379-406	Factor timing & regimes	5
COWY2020	Cederburg et al. (2020)	On the Performance of Volatility-Managed Portfolios	Journal of Financial Economics 138(1):95-117	Factor timing & regimes	3
HKS2020	Haddad, Kozak & Santosh (2020)	Factor Timing	Review of Financial Studies 33(5):1980-2018	Factor timing & regimes	4
LT2020	Lochstoer & Tetlock (2020)	What Drives Anomaly Returns?	Journal of Finance 75(3):1417-1455	Factor timing & regimes	2
NML2018	Nystrup, Madsen & Lindstrom (2018)	Dynamic Portfolio Optimization across Hidden Market Regimes	Quantitative Finance 18(1):83-95	Factor timing & regimes	3
MM2017	Moreira & Muir (2017)	Volatility-Managed Portfolios	Journal of Finance 72(4):1611-1644	Factor timing & regimes	3
DM2016	Daniel & Moskowitz (2016)	Momentum Crashes	Journal of Financial Economics 122(2):221-247	Factor timing & regimes	4
BSC2015	Barroso & Santa-Clara (2015)	Momentum Has Its Moments	Journal of Financial Economics 116(1):111-120	Factor timing & regimes	3
HJTZ2015	Huang et al. (2015)	Investor Sentiment Aligned: A Powerful Predictor of Stock Returns	Review of Financial Studies 28(3):791-837	Factor timing & regimes	2
BAMS2014	Bae, Kim & Mulvey (2014)	Dynamic Asset Allocation for Varied Financial Markets under Regime Switching Framework	European Journal of Operational Research 234(2):450-458	Factor timing & regimes	2
AT2012	Ang & Timmermann (2012)	Regime Changes and Financial Markets	Annual Review of Financial Economics 4:313-337	Factor timing & regimes	3
KPT2012	Kritzman, Page & Turkington (2012)	Regime Shifts: Implications for Dynamic Strategies	Financial Analysts Journal 68(3):22-39	Factor timing & regimes	2
SY2012	Stambaugh, Yu & Yuan (2012)	The Short of It: Investor Sentiment and Anomalies	Journal of Financial Economics 104(2):288-302	Factor timing & regimes	4
RSZ2010	Rapach, Strauss & Zhou (2010)	Out-of-Sample Equity Premium Prediction: Combination Forecasts and Links to the Real...	Review of Financial Studies 23(2):821-862	Factor timing & regimes	4
CT2008	Campbell & Thompson (2008)	Predicting Excess Stock Returns Out of Sample: Can Anything Beat the Historical Average?	Review of Financial Studies 21(4):1509-1531	Factor timing & regimes	2

ID	Citation	Title	Venue	Strand	Rel.
WG2008	Welch & Goyal (2008)	A Comprehensive Look at the Empirical Performance of Equity Premium Prediction	Review of Financial Studies 21(4):1455-1508	Factor timing & regimes	3
GT2007	Guidolin & Timmermann (2007)	Asset Allocation under Multivariate Regime Switching	Journal of Economic Dynamics and Control...	Factor timing & regimes	2
AB2002	Ang & Bekaert (2002)	International Asset Allocation with Regime Shifts	Review of Financial Studies 15(4):1137-1187	Factor timing & regimes	2
HAM1989	Hamilton (1989)	A New Approach to the Economic Analysis of Nonstationary Time Series and the Business...	Econometrica 57(2):357-384	Factor timing & regimes	2
BLACKROCK2025	BlackRock Systematic (2025)	Augmented Investment Management: A systematic framework for designing alpha models	Industry white paper (BlackRock, Nov 2025; for...	Industry	3
VSC2026	Afsharhaji & Li (2026)	The Virtue of Sparsity in Complexity	Working paper (arXiv 2604.17166)	ML asset pricing	4
AIPT2025	Kelly et al. (2025)	APT or AIPT? The Surprising Dominance of Large Factor Models	Working paper (EDHEC-hosted PDF, Oct 2025) [verify...	ML asset pricing	3
BPZ2025	Bryzgalova, Pelger & Zhu (2025)	Forest through the Trees: Building Cross-Sections of Stock Returns	Journal of Finance 80:2447-2506	ML asset pricing	3
CFHH2025	Cong et al. (2025)	Growing the Efficient Frontier on Panel Trees	Journal of Financial Economics 167:104024	ML asset pricing	3
CHSWZ2025	Capponi et al. (2025)	The Nonstationarity-Complexity Tradeoff in Return Prediction	Working paper (arXiv 2512.23596, revised 2026)	ML asset pricing	4
KKMX2025	Kelly et al. (2025)	Artificial Intelligence Asset Pricing Models	Working paper (NBER w33351)	ML asset pricing	3
LLG2025	Chen, Kelly & Malamud (2025)	Limits To (Machine) Learning	Working paper (arXiv 2512.12735)	ML asset pricing	3
LMNS2025	Liao et al. (2025)	The Uncertainty of Machine Learning Predictions in Asset Pricing	Working paper (arXiv 2503.00549)	ML asset pricing	5
SERT2025	Lai (2025)	Asset Pricing in Pre-trained Transformer	Working paper (arXiv 2505.01575)	ML asset pricing	3
WL2025	Wang & Lera (2025)	Meta-Learning for Return Prediction in Shifting Market Regimes	Journal of Financial Markets 79:101042	ML asset pricing	4
CPZ2024	Chen, Pelger & Zhu (2024)	Deep Learning in Asset Pricing	Management Science 70(2):714-750	ML asset pricing	4
KMZ2024	Kelly, Malamud & Zhou (2024)	The Virtue of Complexity in Return Prediction	Journal of Finance 79(1):459-503	ML asset pricing	3
BBCGT2023	Bali et al. (2023)	Option Return Predictability with Machine Learning and Big Data	Review of Financial Studies 36(9):3548-3602	ML asset pricing	2
CFMZ2023	Cakici et al. (2023)	Machine Learning Goes Global: Cross-Sectional Return Predictability in International...	Journal of Economic Dynamics and Control 155:104725	ML asset pricing	3
DA2023	Zhao, Kong & Shen (2023)	DoubleAdapt: A Meta-learning Approach to Incremental Learning for Stock Trend Forecasting	Proceedings of ACM SIGKDD 2023	ML asset pricing	2

ID	Citation	Title	Venue	Strand	Rel.
DKKM2023	Didisheim et al. (2023)	Complexity in Factor Pricing Models	Working paper (NBER w31689; SSRN 4574634)	ML asset pricing	3
HK2023	Hanauer & Kalsbach (2023)	Machine Learning and the Cross-Section of Emerging Market Stock Returns	Emerging Markets Review 55:101022	ML asset pricing	2
JKX2023	Jiang, Kelly & Xiu (2023)	(Re-)Imag(in)ing Price Trends	Journal of Finance 78(6):3193-3249	ML asset pricing	2
KMN2023	Kaniel et al. (2023)	Machine-Learning the Skill of Mutual Fund Managers	Journal of Financial Economics 150(1):94-138	ML asset pricing	1
KMP2023	Kelly, Malamud & Pedersen (2023)	Principal Portfolios	Journal of Finance 78(1):347-387	ML asset pricing	3
KX2023	Kelly & Xiu (2023)	Financial Machine Learning	Foundations and Trends in Finance 13(3-4):205-363	ML asset pricing	4
LLT2023	Lopez-Lira & Tang (2023)	Can ChatGPT Forecast Stock Price Movements? Return Predictability and Large Language...	Working paper (arXiv 2304.07619; SSRN 4412788);...	ML asset pricing	2
VBHL2023	van Binsbergen, Han & Lopez-Lira (2023)	Man versus Machine Learning: The Term Structure of Earnings Expectations and...	Review of Financial Studies 36(6):2361-2396	ML asset pricing	2
GKX2022	Giglio, Kelly & Xiu (2022)	Factor Models, Machine Learning, and Asset Pricing	Annual Review of Financial Economics 14:337-368	ML asset pricing	3
LWZ2022	Leippold, Wang & Zhou (2022)	Machine Learning in the Chinese Stock Market	Journal of Financial Economics 145(2):64-82	ML asset pricing	3
OP2022	Obaid & Pukthuthong (2022)	A Picture Is Worth a Thousand Words: Measuring Investor Sentiment by Combining Machine...	Journal of Financial Economics 144(1):273-297	ML asset pricing	1
BBT2021	Bianchi, Buchner & Tamoni (2021)	Bond Risk Premiums with Machine Learning	Review of Financial Studies 34(2):1046-1089	ML asset pricing	2
GKX2021	Gu, Kelly & Xiu (2021)	Autoencoder Asset Pricing Models	Journal of Econometrics 222(1):429-450	ML asset pricing	3
KMP2021	Kelly, Moskowitz & Pruitt (2021)	Understanding Momentum and Reversal	Journal of Financial Economics 140(3):726-743	ML asset pricing	2
NAG2021	Nagel (2021)	Machine Learning in Asset Pricing	Princeton University Press (book)	ML asset pricing	3
FNW2020	Freyberger, Neuhierl & Weber (2020)	Dissecting Characteristics Nonparametrically	Review of Financial Studies 33(5):2326-2377	ML asset pricing	4
GKX2020	Gu, Kelly & Xiu (2020)	Empirical Asset Pricing via Machine Learning	Review of Financial Studies 33(5):2223-2273	ML asset pricing	5

ID	Citation	Title	Venue	Strand	Rel.
KNS2020	Kozak, Nagel & Santosh (2020)	Shrinking the Cross-Section	Journal of Financial Economics 135(2):271-292	ML asset pricing	4
LP2020	Lettau & Pelger (2020)	Factors That Fit the Time Series and Cross-Section of Stock Returns	Review of Financial Studies 33(5):2274-2325	ML asset pricing	2
CCY2019	Chinco, Clark-Joseph & Ye (2019)	Sparse Signals in the Cross-Section of Returns	Journal of Finance 74(1):449-492	ML asset pricing	2
KPS2019	Kelly, Pruitt & Su (2019)	Characteristics Are Covariances: A Unified Model of Risk and Return	Journal of Financial Economics 134(3):501-524	ML asset pricing	3
KNS2018	Kozak, Nagel & Santosh (2018)	Interpreting Factor Models	Journal of Finance 73(3):1183-1223	ML asset pricing	2
LMR2017	Light, Maslov & Rytchkov (2017)	Aggregation of Information About the Cross Section of Stock Returns: A Latent Variable...	Review of Financial Studies 30(4):1339-1381	ML asset pricing	3
KP2015	Kelly & Pruitt (2015)	The Three-Pass Regression Filter: A New Approach to Forecasting Using Many Predictors	Journal of Econometrics 186(2):294-316	ML asset pricing	2
LEW2015	Lewellen (2015)	The Cross-section of Expected Stock Returns	Critical Finance Review 4(1):1-44	ML asset pricing	5
AZ2020	Apley & Zhu (2020)	Visualizing the Effects of Predictor Variables in Black Box Supervised Learning Models	Journal of the Royal Statistical Society Series B...	ML methods	3
FAL2017	Finn, Abbeel & Levine (2017)	Model-Agnostic Meta-Learning for Fast Adaptation of Deep Networks	Proceedings of ICML 2017 (PMLR 70)	ML methods	2
KE2017	Ke et al. (2017)	LightGBM: A Highly Efficient Gradient Boosting Decision Tree	Advances in Neural Information Processing Systems 30	ML methods	3
LL2017	Lundberg & Lee (2017)	A Unified Approach to Interpreting Model Predictions	Advances in Neural Information Processing Systems 30	ML methods	4
LPB2017	Lakshminarayanan, Pritzel & Blundell (2017)	Simple and Scalable Predictive Uncertainty Estimation Using Deep Ensembles	Advances in Neural Information Processing Systems 30	ML methods	4
VAS2017	Vaswani et al. (2017)	Attention Is All You Need	Advances in Neural Information Processing Systems 30	ML methods	2
CG2016	Chen & Guestrin (2016)	XGBoost: A Scalable Tree Boosting System	Proceedings of ACM SIGKDD 2016	ML methods	3
GG2016	Gal & Ghahramani (2016)	Dropout as a Bayesian Approximation: Representing Model Uncertainty in Deep Learning	Proceedings of ICML 2016 (PMLR 48)	ML methods	3
HTF2009	Hastie, Tibshirani & Friedman (2009)	The Elements of Statistical Learning (2nd ed.)	Springer (book)	ML methods	2

ID	Citation	Title	Venue	Strand	Rel.
RR2007	Rahimi & Recht (2007)	Random Features for Large-Scale Kernel Machines	Advances in Neural Information Processing Systems 20	ML methods	3
ZH2005	Zou & Hastie (2005)	Regularization and Variable Selection via the Elastic Net	Journal of the Royal Statistical Society Series B...	ML methods	2
BRE2001	Breiman (2001)	Random Forests	Machine Learning 45(1):5-32	ML methods	3
FRI2001	Friedman (2001)	Greedy Function Approximation: A Gradient Boosting Machine	Annals of Statistics 29(5):1189-1232	ML methods	3
HS1997	Hochreiter & Schmidhuber (1997)	Long Short-Term Memory	Neural Computation 9(8):1735-1780	ML methods	2
TIB1996	Tibshirani (1996)	Regression Shrinkage and Selection via the Lasso	Journal of the Royal Statistical Society Series B...	ML methods	2
JJNH1991	Jacobs et al. (1991)	Adaptive Mixtures of Local Experts	Neural Computation 3(1):79-87	ML methods	3
JKMP2026	Jensen et al. (2026)	Machine Learning and the Implementable Efficient Frontier	Review of Financial Studies (published online 2026)	ML strategies & implementability	5
LAL2025	Lalwani, Meshram & Jindal (2025)	Empirical Asset Pricing via Machine Learning: The Role of Research Design Choices	European Financial Management 32(3):816-833	ML strategies & implementability	4
CHK2024	Chen, Hanauer & Kalsbach (2024)	Design Choices, Machine Learning, and the Cross-Section of Stock Returns	Working paper (SSRN 5031755)	ML strategies & implementability	5
ACM2023	Avramov, Cheng & Metzker (2023)	Machine Learning vs. Economic Restrictions: Evidence from Stock Return Predictability	Management Science 69(5):2587-2619	ML strategies & implementability	5
AHV2023	Azevedo, Hoegner & Velikov (2023)	The Expected Returns on Machine-Learning Strategies	Working paper (SSRN 4702406); AFA programme	ML strategies & implementability	5
BHHH2023	Blitz et al. (2023)	The Term Structure of Machine Learning Alpha	Journal of Financial Data Science 5(4) 5(4):40-65	ML strategies & implementability	5
LLMSS2021	Leung et al. (2021)	The Promises and Pitfalls of Machine Learning for Predicting Stock Returns	Journal of Financial Data Science 3(2):21-50	ML strategies & implementability	3
TH2021	Tobek & Hronec (2021)	Does It Pay to Follow Anomalies Research? Machine Learning Approach with International...	Journal of Financial Markets 56:100588	ML strategies & implementability	4
RJ2019	Rasekhshafie & Jones (2019)	Machine Learning for Stock Selection	Financial Analysts Journal 75(3):70-88	ML strategies & implementability	4
FK2018	Fischer & Krauss (2018)	Deep Learning with Long Short-Term Memory Networks for Financial Market Predictions	European Journal of Operational Research 270(2):654-669	ML strategies & implementability	2
KDH2017	Krauss, Do & Huck (2017)	Deep Neural Networks, Gradient-Boosted Trees, Random Forests: Statistical Arbitrage on...	European Journal of Operational Research 259(2):689-702	ML strategies & implementability	2
FMY2025	Fan et al. (2025)	Cost-aware Portfolios in a Large Universe of Assets	Working paper (arXiv 2412.11575, revised Aug 2025)	Portfolio construction	3
CPPS2024	Kato (2024)	Conformal Predictive Portfolio Selection	Working paper (arXiv 2410.16333, revised 2025)	Portfolio construction	4

ID	Citation	Title	Venue	Strand	Rel.
BCZ2022	Blanchet, Chen & Zhou (2022)	Distributionally Robust Mean-Variance Portfolio Selection with Wasserstein Distances	Management Science 68(9):6382-6410	Portfolio construction	3
BS2022	Barroso & Saxena (2022)	Lest We Forget: Learn from Out-of-Sample Forecast Errors When Optimizing Portfolios	Review of Financial Studies 35(3):1222-1278	Portfolio construction	4
DMNU2020	DeMiguel et al. (2020)	A Transaction-Cost Perspective on the Multitude of Firm Characteristics	Review of Financial Studies 33(5):2180-2222	Portfolio construction	5
LW2017	Ledoit & Wolf (2017)	Nonlinear Shrinkage of the Covariance Matrix for Portfolio Selection: Markowitz Meets...	Review of Financial Studies 30(12):4349-4388	Portfolio construction	4
LDP2016	Lopez de Prado (2016)	Building Diversified Portfolios that Outperform Out of Sample	Journal of Portfolio Management 42(4):59-69	Portfolio construction	3
GP2013	Garleanu & Pedersen (2013)	Dynamic Trading with Predictable Returns and Transaction Costs	Journal of Finance 68(6):2309-2340	Portfolio construction	5
CSS2012	Ceria, Saxena & Stubbs (2012)	Factor Alignment Problems and Quantitative Portfolio Management	Journal of Portfolio Management 38(2):29-43	Portfolio construction	4
KO2012	Kirby & Ostdiek (2012)	It's All in the Timing: Simple Active Portfolio Strategies that Outperform Naive...	Journal of Financial and Quantitative Analysis...	Portfolio construction	3
DY2010	Delage & Ye (2010)	Distributionally Robust Optimization under Moment Uncertainty with Application to...	Operations Research 58(3):595-612	Portfolio construction	2
MRT2010	Maillard, Roncalli & Teiletche (2010)	The Properties of Equally Weighted Risk Contribution Portfolios	Journal of Portfolio Management 36(4):60-70	Portfolio construction	2
BSCV2009	Brandt, Santa-Clara & Valkanov (2009)	Parametric Portfolio Policies: Exploiting Characteristics in the Cross-Section of...	Review of Financial Studies 22(9):3411-3447	Portfolio construction	4
DGU2009	DeMiguel, Garlappi & Uppal (2009)	Optimal Versus Naïve Diversification: How Inefficient Is the 1/N Portfolio Strategy?	Review of Financial Studies 22(5):1915-1953	Portfolio construction	4
CC2008	Choueifaty & Coignard (2008)	Toward Maximum Diversification	Journal of Portfolio Management 35(1):40-51	Portfolio construction	2
LS2008	Lee & Stefek (2008)	Do Risk Factors Eat Alphas?	Journal of Portfolio Management 34(4):12-25	Portfolio construction	3
KZ2007	Kan & Zhou (2007)	Optimal Portfolio Choice with Parameter Uncertainty	Journal of Financial and Quantitative Analysis...	Portfolio construction	3
LW2004	Ledoit & Wolf (2004)	Honey, I Shrunk the Sample Covariance Matrix	Journal of Portfolio Management 30(4):110-119	Portfolio construction	3
GI2003	Goldfarb & Iyengar (2003)	Robust Portfolio Selection Problems	Mathematics of Operations Research 28(1):1-38	Portfolio construction	3

ID	Citation	Title	Venue	Strand	Rel.
JM2003	Jagannathan & Ma (2003)	Risk Reduction in Large Portfolios: Why Imposing the Wrong Constraints Helps	Journal of Finance 58(4):1651-1683	Portfolio construction	3
CST2002	Clarke, de Silva & Thorley (2002)	Portfolio Constraints and the Fundamental Law of Active Management	Financial Analysts Journal 58(5):48-66	Portfolio construction	4
GK2000	Grinold & Kahn (2000)	Active Portfolio Management (2nd ed.)	McGraw-Hill (book)	Portfolio construction	4
RU2000	Rockafellar & Uryasev (2000)	Optimization of Conditional Value-at-Risk	Journal of Risk 2(3):21-41	Portfolio construction	2
BL1992	Black & Litterman (1992)	Global Portfolio Optimization	Financial Analysts Journal 48(5):28-43	Portfolio construction	3
GRIN1989	Grinold (1989)	The Fundamental Law of Active Management	Journal of Portfolio Management 15(3):30-37	Portfolio construction	4
MICH1989	Michaud (1989)	The Markowitz Optimization Enigma: Is 'Optimized' Optimal?	Financial Analysts Journal 1989(4):43-54	Portfolio construction	3
MARK1952	Markowitz (1952)	Portfolio Selection	Journal of Finance 7(1):77-91	Portfolio construction	3
MENK2024	Menkveld et al. (2024)	Nonstandard Errors	Journal of Finance 79(3):2339-2390	Replication & multiple testing	4
JKP2023	Jensen, Kelly & Pedersen (2023)	Is There a Replication Crisis in Finance?	Journal of Finance 78(5):2465-2518	Replication & multiple testing	5
CHEN2021	Chen (2021)	The Limits of p-Hacking: Some Thought Experiments	Journal of Finance 76(5):2447-2480	Replication & multiple testing	2
GLX2021	Giglio, Liao & Xiu (2021)	Thousands of Alpha Tests	Review of Financial Studies 34(7):3456-3496	Replication & multiple testing	3
FGX2020	Feng, Giglio & Xiu (2020)	Taming the Factor Zoo: A Test of New Factors	Journal of Finance 75(3):1327-1370	Replication & multiple testing	3
HXZ2020	Hou, Xue & Zhang (2020)	Replicating Anomalies	Review of Financial Studies 33(5):2019-2133	Replication & multiple testing	4
GHZ2017	Green, Hand & Zhang (2017)	The Characteristics that Provide Independent Information about Average U.S. Monthly...	Review of Financial Studies 30(12):4389-4436	Replication & multiple testing	3
HAR2017	Harvey (2017)	Presidential Address: The Scientific Outlook in Financial Economics	Journal of Finance 72(4):1399-1440	Replication & multiple testing	4
HLZ2016	Harvey, Liu & Zhu (2016)	... and the Cross-Section of Expected Returns	Review of Financial Studies 29(1):5-68	Replication & multiple testing	5
MP2016	McLean & Pontiff (2016)	Does Academic Research Destroy Stock Return Predictability?	Journal of Finance 71(1):5-32	Replication & multiple testing	4
COC2011	Cochrane (2011)	Presidential Address: Discount Rates	Journal of Finance 66(4):1047-1108	Replication & multiple testing	3

ID	Citation	Title	Venue	Strand	Rel.
SPUR2026	[verify authors] (2026)	Spurious Predictability in Financial Machine Learning	Working paper (arXiv 2604.15531)	Statistical inference	5
AHM2019	Arnett, Harvey & Markowitz (2019)	A Backtesting Protocol in the Era of Machine Learning	Journal of Financial Data Science 1(1):64-74	Statistical inference	5
LDP2018	Lopez de Prado (2018)	Advances in Financial Machine Learning	Wiley (book)	Statistical inference	4
BBLZ2017	Bailey et al. (2017)	The Probability of Backtest Overfitting	Journal of Computational Finance 20(4):39-69	Statistical inference	5
HL2015	Harvey & Liu (2015)	Backtesting	Journal of Portfolio Management 42(1):13-28	Statistical inference	4
BLDP2014	Bailey & Lopez de Prado (2014)	The Deflated Sharpe Ratio: Correcting for Selection Bias, Backtest Overfitting, and...	Journal of Portfolio Management 40(5):94-107	Statistical inference	5
HLN2011	Hansen, Lunde & Nason (2011)	The Model Confidence Set	Econometrica 79(2):453-497	Statistical inference	4
LW2008	Ledoit & Wolf (2008)	Robust Performance Hypothesis Testing with the Sharpe Ratio	Journal of Empirical Finance 15(5):850-859	Statistical inference	5
CW2007	Clark & West (2007)	Approximately Normal Tests for Equal Predictive Accuracy in Nested Models	Journal of Econometrics 138(1):291-311	Statistical inference	4
HAN2005	Hansen (2005)	A Test for Superior Predictive Ability	Journal of Business & Economic Statistics 23(4):365-380	Statistical inference	5
RW2005	Romano & Wolf (2005)	Stepwise Multiple Testing as Formalized Data Snooping	Econometrica 73(4):1237-1282	Statistical inference	4
LO2002	Lo (2002)	The Statistics of Sharpe Ratios	Financial Analysts Journal 58(4):36-52	Statistical inference	4
WHI2000	White (2000)	A Reality Check for Data Snooping	Econometrica 68(5):1097-1126	Statistical inference	5
BP1998	Bai & Perron (1998)	Estimating and Testing Linear Models with Multiple Structural Changes	Econometrica 66(1):47	Statistical inference	3
BH1995	Benjamin i & Hochberg (1995)	Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple...	Journal of the Royal Statistical Society Series B...	Statistical inference	3
DM1995	Diebold & Mariano (1995)	Comparing Predictive Accuracy	Journal of Business & Economic Statistics 13(3):253-263	Statistical inference	4
PR1994	Politis & Romano (1994)	The Stationary Bootstrap	Journal of the American Statistical Association...	Statistical inference	4
NW1987	Newey & West (1987)	A Simple, Positive Semi-Definite, Heteroskedasticity and Autocorrelation Consistent...	Econometrica 55(3):703-708	Statistical inference	4
JK1981	Jobson & Korkie (1981)	Performance Hypothesis Testing with the Sharpe and Treynor Measures	Journal of Finance 36(4):889-908	Statistical inference	3
FM1973	Fama & MacBeth (1973)	Risk, Return, and Equilibrium: Empirical Tests	Journal of Political Economy 81(3):607-636	Statistical inference	3

ID	Citation	Title	Venue	Strand	Rel.
CW2026	Chen & Welch (2026)	What Useful Alphas?	Working paper (arXiv 2607.06502)	TC & implementability	4
AGK2024	Ardia, Guidotti & Kroencke (2024)	Efficient Estimation of Bid-Ask Spreads from Open, High, Low, and Close Prices	Journal of Financial Economics 161:103916	TC & implementability	4
CV2023	Chen & Velikov (2023)	Zeroing In on the Expected Returns of Anomalies	Journal of Financial and Quantitative Analysis...	TC & implementability	5
DNMV2023	Detzel, Novy-Marx & Velikov (2023)	Model Comparison with Transaction Costs	Journal of Finance 78(3):1743-1775	TC & implementability	3
PW2020	Patton & Weller (2020)	What You See Is Not What You Get: The Costs of Trading Market Anomalies	Journal of Financial Economics 137(2):515-549	TC & implementability	3
NMV2019	Novy-Marx & Velikov (2019)	Comparing Cost-Mitigation Techniques	Financial Analysts Journal 75(1):85-102	TC & implementability	3
FIM2018	Frazzini, Israel & Moskowitz (2018)	Trading Costs	Working paper (SSRN 3229719)	TC & implementability	4
AR2017	Abdi & Rinaldo (2017)	A Simple Estimation of Bid-Ask Spreads from Daily Close, High, and Low Prices	Review of Financial Studies 30(12):4437-4480	TC & implementability	3
KYO2016	Kyle & Obizhaeva (2016)	Market Microstructure Invariance: Empirical Hypotheses	Econometrica 84(4):1345-1404	TC & implementability	3
NMV2016	Novy-Marx & Velikov (2016)	A Taxonomy of Anomalies and Their Trading Costs	Review of Financial Studies 29(1):104-147	TC & implementability	5
CST2014	Chordia, Subrahmanyam & Tong (2014)	Have Capital Market Anomalies Attenuated in the Recent Era of High Liquidity and...	Journal of Accounting and Economics 58(1):41-58	TC & implementability	3
CS2012	Corwin & Schultz (2012)	A Simple Way to Estimate Bid-Ask Spreads from Daily High and Low Prices	Journal of Finance 67(2):719-760	TC & implementability	3
HAS2009	Hasbrouck (2009)	Trading Costs and Returns for U.S. Equities: Estimating Effective Costs from Daily Data	Journal of Finance 64(3):1445-1477	TC & implementability	4
ATHL2005	Almgren et al. (2005)	Direct Estimation of Equity Market Impact	Risk 18(7):58-62	TC & implementability	3
KS2004	Korajczyk & Sadka (2004)	Are Momentum Profits Robust to Trading Costs?	Journal of Finance 59(3):1039-1082	TC & implementability	3
LSZ2004	Lesmond, Schill & Zhou (2004)	The Illusory Nature of Momentum Profits	Journal of Financial Economics 71(2):349-380	TC & implementability	2
AMI2002	Amihud (2002)	Illiquidity and Stock Returns: Cross-Section and Time-Series Effects	Journal of Financial Markets 5(1):31-56	TC & implementability	3
AC2001	Almgren & Chriss (2001)	Optimal Execution of Portfolio Transactions	Journal of Risk 3(2):5-39	TC & implementability	3

References

All references cited in this report. DOIs were verified against Crossref or publisher pages where shown; working papers, books and conference papers may lack DOIs. Verification status per entry: 08_References/crossref_verification.csv.

- [verify authors] (2026). Spurious Predictability in Financial Machine Learning. *Working paper (arXiv 2604.15531)*.
- Abdi, F., & Rinaldo, A. (2017). A Simple Estimation of Bid-Ask Spreads from Daily Close, High, and Low Prices. *Review of Financial Studies* 30(12):4437-4480. <https://doi.org/10.1093/rfs/hhx084>
- Afsharhajari, N., & Li, J.Y.-M. (2026). The Virtue of Sparsity in Complexity. *Working paper (arXiv 2604.17166)*.
- Agrawal, A., Amos, B., Barratt, S., Boyd, S., Diamond, S., & Kolter, J.Z. (2019). Differentiable Convex Optimization Layers. *Advances in Neural Information Processing Systems* 32.
- Almgren, R., & Chriss, N. (2001). Optimal Execution of Portfolio Transactions. *Journal of Risk* 3(2):5-39. <https://doi.org/10.21314/jor.2001.041>
- Almgren, R., Thum, C., Hauptmann, E., & Li, H. (2005). Direct Estimation of Equity Market Impact. *Risk* 18(7):58-62.
- Amihud, Y. (2002). Illiquidity and Stock Returns: Cross-Section and Time-Series Effects. *Journal of Financial Markets* 5(1):31-56. [https://doi.org/10.1016/s1386-4181\(01\)00024-6](https://doi.org/10.1016/s1386-4181(01)00024-6)
- Amos, B., & Kolter, J.Z. (2017). OptNet: Differentiable Optimization as a Layer in Neural Networks. *Proceedings of ICML 2017 (PMLR 70)*.
- Ang, A., & Bekaert, G. (2002). International Asset Allocation with Regime Shifts. *Review of Financial Studies* 15(4):1137-1187. <https://doi.org/10.1093/rfs/15.4.1137>
- Ang, A., Hodrick, R.J., Xing, Y., & Zhang, X. (2006). The Cross-Section of Volatility and Expected Returns. *Journal of Finance* 61(1):259-299. <https://doi.org/10.1111/j.1540-6261.2006.00836.x>
- Ang, A., & Timmermann, A. (2012). Regime Changes and Financial Markets. *Annual Review of Financial Economics* 4:313-337. <https://doi.org/10.1146/annurev-financial-110311-101808>
- Apley, D.W., & Zhu, J. (2020). Visualizing the Effects of Predictor Variables in Black Box Supervised Learning Models. *Journal of the Royal Statistical Society Series B* 82(4):1059-1086. <https://doi.org/10.1111/rssb.12377>
- Ardia, D., Guidotti, E., & Kroencke, T.A. (2024). Efficient Estimation of Bid-Ask Spreads from Open, High, Low, and Close Prices. *Journal of Financial Economics* 161:103916. <https://doi.org/10.1016/j.jfineco.2024.103916>
- Arnott, R., Harvey, C.R., & Markowitz, H. (2019). A Backtesting Protocol in the Era of Machine Learning. *Journal of Financial Data Science* 1(1):64-74. <https://doi.org/10.3905/jfds.2019.1.064>
- Asness, C.S., Moskowitz, T.J., & Pedersen, L.H. (2013). Value and Momentum Everywhere. *Journal of Finance* 68(3):929-985. <https://doi.org/10.1111/jofi.12021>
- Avramov, D., Cheng, S., & Metzker, L. (2023). Machine Learning vs. Economic Restrictions: Evidence from Stock Return Predictability. *Management Science* 69(5):2587-2619. <https://doi.org/10.1287/mnsc.2022.4449>
- Azevedo, V., Hoegner, C., & Velikov, M. (2023). The Expected Returns on Machine-Learning Strategies. *Working paper (SSRN 4702406)*; AFA programme. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4702406
- Bae, G.I., Kim, W.C., & Mulvey, J.M. (2014). Dynamic Asset Allocation for Varied Financial Markets under Regime Switching Framework. *European Journal of Operational Research* 234(2):450-458. <https://doi.org/10.1016/j.ejor.2013.03.032>
- Bai, J., & Perron, P. (1998). Estimating and Testing Linear Models with Multiple Structural Changes. *Econometrica* 66(1):47. <https://doi.org/10.2307/2998540>
- Bailey, D.H., & Lopez de Prado, M. (2014). The Deflated Sharpe Ratio: Correcting for Selection Bias, Backtest Overfitting, and Non-Normality. *Journal of Portfolio Management* 40(5):94-107. <https://doi.org/10.3905/jpm.2014.40.5.094>
- Bailey, D.H., Borwein, J., Lopez de Prado, M., & Zhu, Q.J. (2017). The Probability of Backtest Overfitting. *Journal of Computational Finance* 20(4):39-69. <https://doi.org/10.21314/JCF.2016.322>
- Baker, M., & Wurgler, J. (2006). Investor Sentiment and the Cross-Section of Stock Returns. *Journal of Finance* 61(4):1645-1680. <https://doi.org/10.1111/j.1540-6261.2006.00885.x>
- Bali, T.G., Beckmeyer, H., Moerke, M., & Weigert, F. (2023). Option Return Predictability with Machine Learning and Big Data. *Review of Financial Studies* 36(9):3548-3602. <https://doi.org/10.1093/rfs/hhad017>
- Barroso, P., & Santa-Clara, P. (2015). Momentum Has Its Moments. *Journal of Financial Economics* 116(1):111-120. <https://doi.org/10.1016/j.jfineco.2014.11.010>
- Barroso, P., & Saxena, K. (2022). Lest We Forget: Learn from Out-of-Sample Forecast Errors When Optimizing Portfolios. *Review of Financial Studies* 35(3):1222-1278. <https://doi.org/10.1093/rfs/hhbab041>
- Benjamini, Y., & Hochberg, Y. (1995). Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing. *Journal of the Royal Statistical Society Series B* 57(1):289-300. <https://doi.org/10.1111/j.2517-6161.1995.tb02031.x>
- Bianchi, D., Buchner, M., & Tamoni, A. (2021). Bond Risk Premiums with Machine Learning. *Review of Financial Studies* 34(2):1046-1089. <https://doi.org/10.1093/rfs/hhaa062>
- Black, F., & Litterman, R. (1992). Global Portfolio Optimization. *Financial Analysts Journal* 48(5):28-43. <https://doi.org/10.2469/faj.v48.n5.28>

- BlackRock Systematic (2025). Augmented Investment Management: A systematic framework for designing alpha models. *Industry white paper (BlackRock, Nov 2025; for professional investors)*.
<https://www.blackrock.com/gls-download/literature/whitepaper/augmented-investment-management.pdf>
- Blanchet, J., Chen, L., & Zhou, X.Y. (2022). Distributionally Robust Mean-Variance Portfolio Selection with Wasserstein Distances. *Management Science* 68(9):6382-6410. <https://doi.org/10.1287/mnsc.2021.4155>
- Blitz, D., Hanauer, M.X., Hoogteijling, T., & Howard, C. (2023). The Term Structure of Machine Learning Alpha. *Journal of Financial Data Science* 5(4) 5(4):40-65. <https://doi.org/10.3905/jfds.2023.1.135>
- Brandt, M.W., Santa-Clara, P., & Valkanov, R. (2009). Parametric Portfolio Policies: Exploiting Characteristics in the Cross-Section of Equity Returns. *Review of Financial Studies* 22(9):3411-3447. <https://doi.org/10.1093/rfs/hhp003>
- Breiman, L. (2001). Random Forests. *Machine Learning* 45(1):5-32. <https://doi.org/10.1023/a:1010933404324>
- Bryzgalova, S., Pelger, M., & Zhu, J. (2025). Forest through the Trees: Building Cross-Sections of Stock Returns. *Journal of Finance* 80:2447-2506. <https://doi.org/10.1111/jofi.13477>
- Bryzgalova, S., Lerner, S., Lettau, M., & Pelger, M. (2025). Missing Financial Data. *Review of Financial Studies* 38(3):803-882. <https://doi.org/10.1093/rfs/hhae036>
- Butler, A., & Kwon, R.H. (2023). Integrating Prediction in Mean-Variance Portfolio Optimization. *Quantitative Finance* 23(3):429-452. <https://doi.org/10.1080/14697688.2022.2162432>
- Cakici, N., Fieberg, C., Metko, D., & Zaremba, A. (2023). Machine Learning Goes Global: Cross-Sectional Return Predictability in International Stock Markets. *Journal of Economic Dynamics and Control* 155:104725. <https://doi.org/10.1016/j.jedc.2023.104725>
- Campbell, J.Y., & Thompson, S.B. (2008). Predicting Excess Stock Returns Out of Sample: Can Anything Beat the Historical Average?. *Review of Financial Studies* 21(4):1509-1531. <https://doi.org/10.1093/rfs/hhm055>
- Capponi, A., Huang, C., Sidaoui, J.A., Wang, K., & Zou, J. (2025). The Nonstationarity-Complexity Tradeoff in Return Prediction. *Working paper (arXiv 2512.23596, revised 2026)*. <https://arxiv.org/abs/2512.23596>
- Carhart, M.M. (1997). On Persistence in Mutual Fund Performance. *Journal of Finance* 52(1):57-82. <https://doi.org/10.1111/j.1540-6261.1997.tb03808.x>
- Cederburg, S., O'Doherty, M.S., Wang, F., & Yan, X.S. (2020). On the Performance of Volatility-Managed Portfolios. *Journal of Financial Economics* 138(1):95-117. <https://doi.org/10.1016/j.jfineco.2020.04.015>
- Ceria, S., Saxena, A., & Stubbs, R.A. (2012). Factor Alignment Problems and Quantitative Portfolio Management. *Journal of Portfolio Management* 38(2):29-43. <https://doi.org/10.3905/jpm.2012.2012.1.021>
- Chen, T., & Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. *Proceedings of ACM SIGKDD 2016*.
- Chen, A.Y. (2021). The Limits of p-Hacking: Some Thought Experiments. *Journal of Finance* 76(5):2447-2480.
- Chen, A.Y., & Zimmermann, T. (2022). Open Source Cross-Sectional Asset Pricing. *Critical Finance Review* 11(2):207-264. <https://doi.org/10.1561/104.00000112>
- Chen, A.Y., & Velikov, M. (2023). Zeroing In on the Expected Returns of Anomalies. *Journal of Financial and Quantitative Analysis* 58(3):968-1004. <https://doi.org/10.1017/S0022109022000874>
- Chen, L., Pelger, M., & Zhu, J. (2024). Deep Learning in Asset Pricing. *Management Science* 70(2):714-750. <https://doi.org/10.1287/mnsc.2023.4695>
- Chen, M., Hanauer, M.X., & Kalsbach, T. (2024). Design Choices, Machine Learning, and the Cross-Section of Stock Returns. *Working paper (SSRN 5031755)*. <https://ssrn.com/abstract=5031755>
- Chen, Z., Kelly, B., & Malamud, S. (2025). Limits To (Machine) Learning. *Working paper (arXiv 2512.12735)*.
- Chen, A.Y., & Welch, I. (2026). What Useful Alphas?. *Working paper (arXiv 2607.06502)*.
- Chinco, A., Clark-Joseph, A.D., & Ye, M. (2019). Sparse Signals in the Cross-Section of Returns. *Journal of Finance* 74(1):449-492. <https://doi.org/10.1111/jofi.12733>
- Chordia, T., Subrahmanyam, A., & Tong, Q. (2014). Have Capital Market Anomalies Attenuated in the Recent Era of High Liquidity and Trading Activity?. *Journal of Accounting and Economics* 58(1):41-58. <https://doi.org/10.1016/j.jacceco.2014.06.001>
- Choueifaty, Y., & Coignard, Y. (2008). Toward Maximum Diversification. *Journal of Portfolio Management* 35(1):40-51. <https://doi.org/10.3905/jpm.2008.35.1.40>
- Clark, T.E., & West, K.D. (2007). Approximately Normal Tests for Equal Predictive Accuracy in Nested Models. *Journal of Econometrics* 138(1):291-311. <https://doi.org/10.1016/j.jeconom.2006.05.023>
- Clarke, R., de Silva, H., & Thorley, S. (2002). Portfolio Constraints and the Fundamental Law of Active Management. *Financial Analysts Journal* 58(5):48-66. <https://doi.org/10.2469/faj.v58.n5.2468>
- Cochrane, J.H. (2011). Presidential Address: Discount Rates. *Journal of Finance* 66(4):1047-1108. <https://doi.org/10.1111/j.1540-6261.2011.01671.x>
- Cohen, L., & Frazzini, A. (2008). Economic Links and Predictable Returns. *Journal of Finance* 63(4):1977-2011. <https://doi.org/10.1111/j.1540-6261.2008.01379.x>
- Collin-Dufresne, P., Daniel, K., & Saglam, M. (2020). Liquidity Regimes and Optimal Dynamic Asset Allocation. *Journal of Financial Economics* 136(2):379-406. <https://doi.org/10.1016/j.jfineco.2019.09.011>
- Cong, L.W., Tang, K., Wang, J., & Zhang, Y. (2021). AlphaPortfolio: Direct Construction through Deep Reinforcement Learning and Interpretable AI. *Working paper (SSRN 3554486; NBER w35195)*.

- Cong, L.W., Feng, G., He, J., & He, X. (2025). Growing the Efficient Frontier on Panel Trees. *Journal of Financial Economics* 167:104024. <https://doi.org/10.1016/j.jfineco.2025.104024>
- Corwin, S.A., & Schultz, P. (2012). A Simple Way to Estimate Bid-Ask Spreads from Daily High and Low Prices. *Journal of Finance* 67(2):719-760. <https://doi.org/10.1111/j.1540-6261.2012.01729.x>
- Costa, G., & Iyengar, G.N. (2023). Distributionally Robust End-to-End Portfolio Construction. *Quantitative Finance* 23(10):1465-1482. <https://doi.org/10.1080/14697688.2023.2236148>
- Daniel, K., & Moskowitz, T.J. (2016). Momentum Crashes. *Journal of Financial Economics* 122(2):221-247. <https://doi.org/10.1016/j.jfineco.2015.12.002>
- Delage, E., & Ye, Y. (2010). Distributionally Robust Optimization under Moment Uncertainty with Application to Data-Driven Problems. *Operations Research* 58(3):595-612. <https://doi.org/10.1287/opre.1090.0741>
- DeMiguel, V., Garlappi, L., & Uppal, R. (2009). Optimal Versus Naive Diversification: How Inefficient Is the 1/N Portfolio Strategy?. *Review of Financial Studies* 22(5):1915-1953. <https://doi.org/10.1093/rfs/hhm075>
- DeMiguel, V., Martin-Utrera, A., Nogales, F.J., & Uppal, R. (2020). A Transaction-Cost Perspective on the Multitude of Firm Characteristics. *Review of Financial Studies* 33(5):2180-2222. <https://doi.org/10.1093/rfs/hhzo85>
- DeMiguel, V., Martin-Utrera, A., & Uppal, R. (2024). A Multifactor Perspective on Volatility-Managed Portfolios. *Journal of Finance* 79:3859-3891. <https://doi.org/10.1111/jofi.13395>
- Detzel, A., Novy-Marx, R., & Velikov, M. (2023). Model Comparison with Transaction Costs. *Journal of Finance* 78(3):1743-1775. <https://doi.org/10.1111/jofi.13225>
- Didisheim, A., Ke, S., Kelly, B., & Malamud, S. (2023). Complexity in Factor Pricing Models. *Working paper (NBER w31689; SSRN 4574634)*.
- Diebold, F.X., & Mariano, R.S. (1995). Comparing Predictive Accuracy. *Journal of Business & Economic Statistics* 13(3):253-263. <https://doi.org/10.1080/07350015.1995.10524599>
- Dong, X., Li, Y., Rapach, D.E., & Zhou, G. (2022). Anomalies and the Expected Market Return. *Journal of Finance* 77(1):639-681. <https://doi.org/10.1111/jofi.13099>
- Donti, P.L., Amos, B., & Kolter, J.Z. (2017). Task-based End-to-end Model Learning in Stochastic Optimization. *Advances in Neural Information Processing Systems* 30.
- Ehsani, S., & Linnainmaa, J.T. (2022). Factor Momentum and the Momentum Factor. *Journal of Finance* 77(3):1877-1919. <https://doi.org/10.1111/jofi.13131>
- Elmachtoub, A.N., & Grigas, P. (2022). Smart 'Predict, then Optimize'. *Management Science* 68(1):9-26. <https://doi.org/10.1287/mnsc.2020.3922>
- Fama, E.F., & MacBeth, J.D. (1973). Risk, Return, and Equilibrium: Empirical Tests. *Journal of Political Economy* 81(3):607-636. <https://doi.org/10.1086/260061>
- Fama, E.F., & French, K.R. (1993). Common Risk Factors in the Returns on Stocks and Bonds. *Journal of Financial Economics* 33(1):3-56. [https://doi.org/10.1016/0304-405x\(93\)90023-5](https://doi.org/10.1016/0304-405x(93)90023-5)
- Fama, E.F., & French, K.R. (2015). A Five-Factor Asset Pricing Model. *Journal of Financial Economics* 116(1):1-22. <https://doi.org/10.1016/j.jfineco.2014.10.010>
- Fan, Q., Medeiros, M.C., Yang, H., & Yang, S. (2025). Cost-aware Portfolios in a Large Universe of Assets. *Working paper (arXiv 2412.11575, revised Aug 2025)*.
- Feng, G., Giglio, S., & Xiu, D. (2020). Taming the Factor Zoo: A Test of New Factors. *Journal of Finance* 75(3):1327-1370. <https://doi.org/10.1111/jofi.12883>
- Finn, C., Abbeel, P., & Levine, S. (2017). Model-Agnostic Meta-Learning for Fast Adaptation of Deep Networks. *Proceedings of ICML 2017 (PMLR 70)*.
- Fischer, T., & Krauss, C. (2018). Deep Learning with Long Short-Term Memory Networks for Financial Market Predictions. *European Journal of Operational Research* 270(2):654-669. <https://doi.org/10.1016/j.ejor.2017.11.054>
- Frazzini, A., & Pedersen, L.H. (2014). Betting Against Beta. *Journal of Financial Economics* 111(1):1-25. <https://doi.org/10.1016/j.jfineco.2013.10.005>
- Frazzini, A., Israel, R., & Moskowitz, T.J. (2018). Trading Costs. *Working paper (SSRN 3229719)*. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3229719
- Freyberger, J., Neuhierl, A., & Weber, M. (2020). Dissecting Characteristics Nonparametrically. *Review of Financial Studies* 33(5):2326-2377. <https://doi.org/10.1093/rfs/hhz123>
- Freyberger, J., Hoppner, B., Neuhierl, A., & Weber, M. (2025). Missing Data in Asset Pricing Panels. *Review of Financial Studies* 38(3):760-802. <https://doi.org/10.1093/rfs/hhae003>
- Friedman, J.H. (2001). Greedy Function Approximation: A Gradient Boosting Machine. *Annals of Statistics* 29(5):1189-1232. <https://doi.org/10.1214/aos/1013203451>
- Gal, Y., & Ghahramani, Z. (2016). Dropout as a Bayesian Approximation: Representing Model Uncertainty in Deep Learning. *Proceedings of ICML 2016 (PMLR 48)*.
- Garleanu, N., & Pedersen, L.H. (2013). Dynamic Trading with Predictable Returns and Transaction Costs. *Journal of Finance* 68(6):2309-2340. <https://doi.org/10.1111/jofi.12080>

- Giglio, S., Liao, Y., & Xiu, D. (2021). Thousands of Alpha Tests. *Review of Financial Studies* 34(7):3456-3496. <https://doi.org/10.1093/rfs/hhaa111>
- Giglio, S., Kelly, B., & Xiu, D. (2022). Factor Models, Machine Learning, and Asset Pricing. *Annual Review of Financial Economics* 14:337-368. <https://doi.org/10.1146/annurev-financial-101521-104735>
- Goldfarb, D., & Iyengar, G. (2003). Robust Portfolio Selection Problems. *Mathematics of Operations Research* 28(1):1-38. <https://doi.org/10.1287/moor.28.1.1.14260>
- Green, J., Hand, J.R.M., & Zhang, X.F. (2017). The Characteristics that Provide Independent Information about Average U.S. Monthly Stock Returns. *Review of Financial Studies* 30(12):4389-4436. <https://doi.org/10.1093/rfs/hhx019>
- Grinold, R.C. (1989). The Fundamental Law of Active Management. *Journal of Portfolio Management* 15(3):30-37. <https://doi.org/10.3905/jpm.1989.409211>
- Grinold, R.C., & Kahn, R.N. (2000). *Active Portfolio Management* (2nd ed.). McGraw-Hill (book).
- Gu, S., Kelly, B., & Xiu, D. (2020). Empirical Asset Pricing via Machine Learning. *Review of Financial Studies* 33(5):2223-2273. <https://doi.org/10.1093/rfs/hhaa009>
- Gu, S., Kelly, B., & Xiu, D. (2021). Autoencoder Asset Pricing Models. *Journal of Econometrics* 222(1):429-450. <https://doi.org/10.1016/j.jeconom.2020.07.009>
- Guidolin, M., & Timmermann, A. (2007). Asset Allocation under Multivariate Regime Switching. *Journal of Economic Dynamics and Control* 31(11):3503-3544. <https://doi.org/10.1016/j.jedc.2006.12.004>
- Haddad, V., Kozak, S., & Santosh, S. (2020). Factor Timing. *Review of Financial Studies* 33(5):1980-2018. <https://doi.org/10.1093/rfs/hhaa017>
- Hamilton, J.D. (1989). A New Approach to the Economic Analysis of Nonstationary Time Series and the Business Cycle. *Econometrica* 57(2):357-384. <https://doi.org/10.2307/1912559>
- Hanauer, M.X., & Kalsbach, T. (2023). Machine Learning and the Cross-Section of Emerging Market Stock Returns. *Emerging Markets Review* 55:101022. <https://doi.org/10.1016/j.ememar.2023.101022>
- Hansen, P.R. (2005). A Test for Superior Predictive Ability. *Journal of Business & Economic Statistics* 23(4):365-380. <https://doi.org/10.1198/073500105000000063>
- Hansen, P.R., Lunde, A., & Nason, J.M. (2011). The Model Confidence Set. *Econometrica* 79(2):453-497. <https://doi.org/10.3982/ecta5771>
- Harvey, C.R., & Liu, Y. (2015). Backtesting. *Journal of Portfolio Management* 42(1):13-28. <https://doi.org/10.3905/jpm.2015.42.1.013>
- Harvey, C.R., Liu, Y., & Zhu, H. (2016). ... and the Cross-Section of Expected Returns. *Review of Financial Studies* 29(1):5-68. <https://doi.org/10.1093/rfs/hhv059>
- Harvey, C.R. (2017). Presidential Address: The Scientific Outlook in Financial Economics. *Journal of Finance* 72(4):1399-1440. <https://doi.org/10.1111/jofi.12530>
- Hasbrouck, J. (2009). Trading Costs and Returns for U.S. Equities: Estimating Effective Costs from Daily Data. *Journal of Finance* 64(3):1445-1477. <https://doi.org/10.1111/j.1540-6261.2009.01469.x>
- Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The Elements of Statistical Learning* (2nd ed.). Springer (book).
- Hochreiter, S., & Schmidhuber, J. (1997). Long Short-Term Memory. *Neural Computation* 9(8):1735-1780. <https://doi.org/10.1162/neco.1997.9.8.1735>
- Hou, K., Xue, C., & Zhang, L. (2015). Digesting Anomalies: An Investment Approach. *Review of Financial Studies* 28(3):650-705. <https://doi.org/10.1093/rfs/hhu068>
- Hou, K., Xue, C., & Zhang, L. (2020). Replicating Anomalies. *Review of Financial Studies* 33(5):2019-2133. <https://doi.org/10.1093/rfs/hhy131>
- Huang, D., Jiang, F., Tu, J., & Zhou, G. (2015). Investor Sentiment Aligned: A Powerful Predictor of Stock Returns. *Review of Financial Studies* 28(3):791-837. <https://doi.org/10.1093/rfs/hhu080>
- Jacobs, R.A., Jordan, M.I., Nowlan, S.J., & Hinton, G.E. (1991). Adaptive Mixtures of Local Experts. *Neural Computation* 3(1):79-87. <https://doi.org/10.1162/neco.1991.3.1.79>
- Jagannathan, R., & Ma, T. (2003). Risk Reduction in Large Portfolios: Why Imposing the Wrong Constraints Helps. *Journal of Finance* 58(4):1651-1683. <https://doi.org/10.1111/1540-6261.00580>
- Jensen, T.I., Kelly, B., & Pedersen, L.H. (2023). Is There a Replication Crisis in Finance?. *Journal of Finance* 78(5):2465-2518. <https://doi.org/10.1111/jofi.13249>
- Jensen, T.I., Kelly, B., Malamud, S., & Pedersen, L.H. (2026). Machine Learning and the Implementable Efficient Frontier. *Review of Financial Studies* (published online 2026). <https://doi.org/10.1093/rfs/hhago22>
- Jiang, J., Kelly, B., & Xiu, D. (2023). (Re-)Imag(in)ing Price Trends. *Journal of Finance* 78(6):3193-3249. <https://doi.org/10.1111/jofi.13268>
- Jobson, J.D., & Korkie, B.M. (1981). Performance Hypothesis Testing with the Sharpe and Treynor Measures. *Journal of Finance* 36(4):889-908. <https://doi.org/10.1111/j.1540-6261.1981.tb04891.x>
- Kagkadis, A., Nolte, I., Nolte, S., & Vasilas, N. (2024). Factor Timing with Portfolio Characteristics. *Review of Asset Pricing Studies* 14(1):84-118. <https://doi.org/10.1093/rapstu/raado10>

- Kan, R., & Zhou, G. (2007). Optimal Portfolio Choice with Parameter Uncertainty. *Journal of Financial and Quantitative Analysis* 42(3):621-656. <https://doi.org/10.1017/S0022109000004129>
- Kaniel, R., Lin, Z., Pelger, M., & Van Nieuwerburgh, S. (2023). Machine-Learning the Skill of Mutual Fund Managers. *Journal of Financial Economics* 150(1):94-138. <https://doi.org/10.1016/j.jfineco.2023.07.004>
- Kato, M. (2024). Conformal Predictive Portfolio Selection. *Working paper (arXiv 2410.16333, revised 2025)*.
- Ke, G., et al. (2017). LightGBM: A Highly Efficient Gradient Boosting Decision Tree. *Advances in Neural Information Processing Systems* 30.
- Kelly, B., & Pruitt, S. (2015). The Three-Pass Regression Filter: A New Approach to Forecasting Using Many Predictors. *Journal of Econometrics* 186(2):294-316. <https://doi.org/10.1016/j.jeconom.2015.02.011>
- Kelly, B., Pruitt, S., & Su, Y. (2019). Characteristics Are Covariances: A Unified Model of Risk and Return. *Journal of Financial Economics* 134(3):501-524. <https://doi.org/10.1016/j.jfineco.2019.05.001>
- Kelly, B., Moskowitz, T.J., & Pruitt, S. (2021). Understanding Momentum and Reversal. *Journal of Financial Economics* 140(3):726-743. <https://doi.org/10.1016/j.jfineco.2020.06.024>
- Kelly, B., & Xiu, D. (2023). Financial Machine Learning. *Foundations and Trends in Finance* 13(3-4):205-363. <https://doi.org/10.1561/5000000064>
- Kelly, B., Malamud, S., & Pedersen, L.H. (2023). Principal Portfolios. *Journal of Finance* 78(1):347-387. <https://doi.org/10.1111/jofi.13199>
- Kelly, B., Malamud, S., & Zhou, K. (2024). The Virtue of Complexity in Return Prediction. *Journal of Finance* 79(1):459-503. <https://doi.org/10.1111/jofi.13298>
- Kelly, B., et al. (2025). APT or AIPT? The Surprising Dominance of Large Factor Models. *Working paper (EDHEC-hosted PDF, Oct 2025) [verify authors and venue]*.
- Kelly, B., Kuznetsov, B., Malamud, S., & Xu, T.A. (2025). Artificial Intelligence Asset Pricing Models. *Working paper (NBER w33351)*.
- Kirby, C., & Ostdiek, B. (2012). It's All in the Timing: Simple Active Portfolio Strategies that Outperform Naive Diversification. *Journal of Financial and Quantitative Analysis* 47(2):437-467. <https://doi.org/10.1017/S0022109012000117>
- Korajczyk, R.A., & Sadka, R. (2004). Are Momentum Profits Robust to Trading Costs?. *Journal of Finance* 59(3):1039-1082. <https://doi.org/10.1111/j.1540-6261.2004.00656.x>
- Kozak, S., Nagel, S., & Santosh, S. (2018). Interpreting Factor Models. *Journal of Finance* 73(3):1183-1223. <https://doi.org/10.1111/jofi.12612>
- Kozak, S., Nagel, S., & Santosh, S. (2020). Shrinking the Cross-Section. *Journal of Financial Economics* 135(2):271-292. <https://doi.org/10.1016/j.jfineco.2019.06.008>
- Krauss, C., Do, X.A., & Huck, N. (2017). Deep Neural Networks, Gradient-Boosted Trees, Random Forests: Statistical Arbitrage on the S&P 500. *European Journal of Operational Research* 259(2):689-702. <https://doi.org/10.1016/j.ejor.2016.10.031>
- Kritzman, M., Page, S., & Turkington, D. (2012). Regime Shifts: Implications for Dynamic Strategies. *Financial Analysts Journal* 68(3):22-39. <https://doi.org/10.2469/faj.v68.n3.3>
- Kyle, A.S., & Obizhaeva, A.A. (2016). Market Microstructure Invariance: Empirical Hypotheses. *Econometrica* 84(4):1345-1404. <https://doi.org/10.3982/ecta10486>
- Lai, S. (2025). Asset Pricing in Pre-trained Transformer. *Working paper (arXiv 2505.01575)*.
- Lakshminarayanan, B., Pritzel, A., & Blundell, C. (2017). Simple and Scalable Predictive Uncertainty Estimation Using Deep Ensembles. *Advances in Neural Information Processing Systems* 30.
- Lalwani, V., Meshram, V., & Jindal, V. (2025). Empirical Asset Pricing via Machine Learning: The Role of Research Design Choices. *European Financial Management* 32(3):816-833. <https://doi.org/10.1111/eufm.70033>
- Ledoit, O., & Wolf, M. (2004). Honey, I Shrunk the Sample Covariance Matrix. *Journal of Portfolio Management* 30(4):110-119. <https://doi.org/10.3905/jpm.2004.110>
- Ledoit, O., & Wolf, M. (2008). Robust Performance Hypothesis Testing with the Sharpe Ratio. *Journal of Empirical Finance* 15(5):850-859. <https://doi.org/10.1016/j.jempfin.2008.03.002>
- Ledoit, O., & Wolf, M. (2017). Nonlinear Shrinkage of the Covariance Matrix for Portfolio Selection: Markowitz Meets Goldilocks. *Review of Financial Studies* 30(12):4349-4388. <https://doi.org/10.1093/rfs/hhx052>
- Lee, J.-H., & Stefek, D. (2008). Do Risk Factors Eat Alphas?. *Journal of Portfolio Management* 34(4):12-25. <https://doi.org/10.3905/jpm.2008.709976>
- Leippold, M., Wang, Q., & Zhou, W. (2022). Machine Learning in the Chinese Stock Market. *Journal of Financial Economics* 145(2):64-82. <https://doi.org/10.1016/j.jfineco.2021.08.017>
- Lesmond, D.A., Schill, M.J., & Zhou, C. (2004). The Illusory Nature of Momentum Profits. *Journal of Financial Economics* 71(2):349-380. [https://doi.org/10.1016/S0304-405X\(03\)00206-X](https://doi.org/10.1016/S0304-405X(03)00206-X)
- Lettau, M., & Pelger, M. (2020). Factors That Fit the Time Series and Cross-Section of Stock Returns. *Review of Financial Studies* 33(5):2274-2325. <https://doi.org/10.1093/rfs/hhaz020>
- Leung, E., Lohre, H., Mischlich, D., Shea, Y., & Stroh, M. (2021). The Promises and Pitfalls of Machine Learning for Predicting Stock Returns. *Journal of Financial Data Science* 3(2):21-50. <https://doi.org/10.3905/jfds.2021.1.062>

- Lewellen, J. (2015). The Cross-section of Expected Stock Returns. *Critical Finance Review* 4(1):1-44. <https://doi.org/10.1561/104.00000024>
- Liao, Y., Ma, X., Neuhierl, A., & Schilling, L. (2025). The Uncertainty of Machine Learning Predictions in Asset Pricing. *Working paper (arXiv 2503.00549)*. <https://arxiv.org/abs/2503.00549>
- Light, N., Maslov, D., & Rytchkov, O. (2017). Aggregation of Information About the Cross Section of Stock Returns: A Latent Variable Approach. *Review of Financial Studies* 30(4):1339-1381. <https://doi.org/10.1093/rfs/hhw102>
- Lim, B., Zohren, S., & Roberts, S. (2019). Enhancing Time-Series Momentum Strategies Using Deep Neural Networks. *Journal of Financial Data Science* 1(4):19-38. <https://doi.org/10.3905/jfds.2019.1.015>
- Lo, A.W. (2002). The Statistics of Sharpe Ratios. *Financial Analysts Journal* 58(4):36-52. <https://doi.org/10.2469/faj.v58.n4.2453>
- Lochstoer, L.A., & Tetlock, P.C. (2020). What Drives Anomaly Returns?. *Journal of Finance* 75(3):1417-1455. <https://doi.org/10.1111/jofi.12876>
- Lopez de Prado, M. (2016). Building Diversified Portfolios that Outperform Out of Sample. *Journal of Portfolio Management* 42(4):59-69. <https://doi.org/10.3905/jpm.2016.42.4.059>
- Lopez de Prado, M. (2018). *Advances in Financial Machine Learning*. Wiley (book).
- Lopez-Lira, A., & Tang, Y. (2023). Can ChatGPT Forecast Stock Price Movements? Return Predictability and Large Language Models. *Working paper (arXiv 2304.07619; SSRN 4412788)*; reported forthcoming in *JFE* [verify].
- Lundberg, S.M., & Lee, S.-I. (2017). A Unified Approach to Interpreting Model Predictions. *Advances in Neural Information Processing Systems* 30.
- Maillard, S., Roncalli, T., & Teiletche, J. (2010). The Properties of Equally Weighted Risk Contribution Portfolios. *Journal of Portfolio Management* 36(4):60-70. <https://doi.org/10.3905/jpm.2010.36.4.060>
- Markowitz, H. (1952). Portfolio Selection. *Journal of Finance* 7(1):77-91. <https://doi.org/10.1111/j.1540-6261.1952.tb01525.x>
- McLean, R.D., & Pontiff, J. (2016). Does Academic Research Destroy Stock Return Predictability?. *Journal of Finance* 71(1):5-32. <https://doi.org/10.1111/jofi.12365>
- Menkveld, A.J., et al. (2024). Nonstandard Errors. *Journal of Finance* 79(3):2339-2390.
- Michaud, R.O. (1989). The Markowitz Optimization Enigma: Is 'Optimized' Optimal?. *Financial Analysts Journal* 1989(4):43-54. <https://doi.org/10.2469/cp.v1989.n4.6>
- Moody, J., & Saffell, M. (2001). Learning to Trade via Direct Reinforcement. *IEEE Transactions on Neural Networks* 12(4):875-889. <https://doi.org/10.1109/72.935097>
- Moreira, A., & Muir, T. (2017). Volatility-Managed Portfolios. *Journal of Finance* 72(4):1611-1644. <https://doi.org/10.1111/jofi.12513>
- Muller, S., & Preissler, F. (2026). In Good and in Bad Times? The Relation between Anomaly Returns and Market States. *Journal of Banking & Finance* 190:107746. <https://doi.org/10.1016/j.jbankfin.2026.107746>
- Nagel, S. (2021). *Machine Learning in Asset Pricing*. Princeton University Press (book).
- Neuhierl, A., Randl, O., Reschenhofer, C., & Zechner, J. (2024). Timing the Factor Zoo. *Working paper (SSRN 4376898)*; AFA/AEA conference programmes. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4376898
- Newey, W.K., & West, K.D. (1987). A Simple, Positive Semi-Definite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix. *Econometrica* 55(3):703-708. <https://doi.org/10.2307/1913610>
- Novy-Marx, R., & Velikov, M. (2016). A Taxonomy of Anomalies and Their Trading Costs. *Review of Financial Studies* 29(1):104-147. <https://doi.org/10.1093/rfs/hhv063>
- Novy-Marx, R., & Velikov, M. (2019). Comparing Cost-Mitigation Techniques. *Financial Analysts Journal* 75(1):85-102. <https://doi.org/10.1080/0015198x.2018.1547057>
- Nystrup, P., Madsen, H., & Lindstrom, E. (2018). Dynamic Portfolio Optimization across Hidden Market Regimes. *Quantitative Finance* 18(1):83-95. <https://doi.org/10.1080/14697688.2017.1342857>
- Obaid, K., & Pukthuanthong, K. (2022). A Picture Is Worth a Thousand Words: Measuring Investor Sentiment by Combining Machine Learning and Photos from News. *Journal of Financial Economics* 144(1):273-297. <https://doi.org/10.1016/j.jfineco.2021.06.002>
- Pastor, L., & Stambaugh, R.F. (2003). Liquidity Risk and Expected Stock Returns. *Journal of Political Economy* 111(3):642-685. <https://doi.org/10.1086/374184>
- Patton, A.J., & Weller, B.M. (2020). What You See Is Not What You Get: The Costs of Trading Market Anomalies. *Journal of Financial Economics* 137(2):515-549. <https://doi.org/10.1016/j.jfineco.2020.02.012>
- Politis, D.N., & Romano, J.P. (1994). The Stationary Bootstrap. *Journal of the American Statistical Association* 89(428):1303-1313. <https://doi.org/10.1080/01621459.1994.10476870>
- Rahimi, A., & Recht, B. (2007). Random Features for Large-Scale Kernel Machines. *Advances in Neural Information Processing Systems* 20.
- Rapach, D.E., Strauss, J.K., & Zhou, G. (2010). Out-of-Sample Equity Premium Prediction: Combination Forecasts and Links to the Real Economy. *Review of Financial Studies* 23(2):821-862. <https://doi.org/10.1093/rfs/hhp063>
- Rasekhschaffe, K.C., & Jones, R.C. (2019). Machine Learning for Stock Selection. *Financial Analysts Journal* 75(3):70-88. <https://doi.org/10.1080/0015198X.2019.1596678>

- Rockafellar, R.T., & Uryasev, S. (2000). Optimization of Conditional Value-at-Risk. *Journal of Risk* 2(3):21-41. <https://doi.org/10.21314/jor.2000.038>
- Romano, J.P., & Wolf, M. (2005). Stepwise Multiple Testing as Formalized Data Snooping. *Econometrica* 73(4):1237-1282. <https://doi.org/10.1111/j.1468-0262.2005.00615.x>
- Shu, Y., Yu, C., & Mulvey, J.M. (2025). Dynamic Asset Allocation with Asset-Specific Regime Forecasts. *Annals of Operations Research* 346(1):285-318. <https://doi.org/10.1007/s10479-024-06266-0>
- Shumway, T. (1997). The Delisting Bias in CRSP Data. *Journal of Finance* 52(1):327-340. <https://doi.org/10.1111/j.1540-6261.1997.tb03818.x>
- Simon, F., Weibels, S., & Zimmermann, T. (2026). Deep Parametric Portfolio Policies. *Management Science* (2026; Crossref record) - earlier CFR Working Paper 23-01 / SSRN 4150292. <https://doi.org/10.1287/mnsc.2025.00721>
- Stambaugh, R.F., Yu, J., & Yuan, Y. (2012). The Short of It: Investor Sentiment and Anomalies. *Journal of Financial Economics* 104(2):288-302. <https://doi.org/10.1016/j.jfineco.2011.12.001>
- Stambaugh, R.F., & Yuan, Y. (2017). Mispricing Factors. *Review of Financial Studies* 30(4):1270-1315. <https://doi.org/10.1093/rfs/hhw107>
- Tibshirani, R. (1996). Regression Shrinkage and Selection via the Lasso. *Journal of the Royal Statistical Society Series B* 58(1):267-288. <https://doi.org/10.1111/j.2517-6161.1996.tb02080.x>
- Tobek, O., & Hronec, M. (2021). Does It Pay to Follow Anomalies Research? Machine Learning Approach with International Evidence. *Journal of Financial Markets* 56:100588. <https://doi.org/10.1016/j.finmar.2020.100588>
- Uysal, A.S., Li, X., & Mulvey, J.M. (2024). End-to-End Risk Budgeting Portfolio Optimization with Neural Networks. *Annals of Operations Research* 339(1-2):397-426. <https://doi.org/10.1007/s10479-023-05539-4>
- van Binsbergen, J.H., Han, X., & Lopez-Lira, A. (2023). Man versus Machine Learning: The Term Structure of Earnings Expectations and Conditional Biases. *Review of Financial Studies* 36(6):2361-2396. <https://doi.org/10.1093/rfs/hhac085>
- Vaswani, A., et al. (2017). Attention Is All You Need. *Advances in Neural Information Processing Systems* 30.
- Wang, Y., & Lera, S.C. (2025). Meta-Learning for Return Prediction in Shifting Market Regimes. *Journal of Financial Markets* 79:101042. <https://doi.org/10.1016/j.finmar.2025.101042>
- Wang, Y., & Hasuikie, T. (2026). Decision-Induced Ranking Explains Prediction Inflation and Excessive Turnover in SPO-Based Portfolio Optimization. *Working paper (arXiv 2605.01176)*.
- Welch, I., & Goyal, A. (2008). A Comprehensive Look at the Empirical Performance of Equity Premium Prediction. *Review of Financial Studies* 21(4):1455-1508. <https://doi.org/10.1093/rfs/hhm014>
- White, H. (2000). A Reality Check for Data Snooping. *Econometrica* 68(5):1097-1126. <https://doi.org/10.1111/1468-0262.00152>
- Wood, K., Giegerich, S., Roberts, S., & Zohren, S. (2022). Trading with the Momentum Transformer: An Intelligent and Interpretable Architecture. *arXiv 2112.08534 [verify journal version]*.
- Zhang, Z., Zohren, S., & Roberts, S. (2020). Deep Learning for Portfolio Optimization. *Journal of Financial Data Science* 2(4):8-20. <https://doi.org/10.3905/jfds.2020.1.042>
- Zhang, J. (2025). FR-LUX: Friction-Aware, Regime-Conditioned Policy Optimization for Implementable Portfolio Management. *Working paper (arXiv 2510.02986)*.
- Zhao, L., Kong, S., & Shen, Y. (2023). DoubleAdapt: A Meta-learning Approach to Incremental Learning for Stock Trend Forecasting. *Proceedings of ACM SIGKDD 2023*. <https://doi.org/10.1145/3580305.3599315>
- Zou, H., & Hastie, T. (2005). Regularization and Variable Selection via the Elastic Net. *Journal of the Royal Statistical Society Series B* 67(2):301-320. <https://doi.org/10.1111/j.1467-9868.2005.00503.x>